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Prehospital risk assessment in suspected Acute Coronary Syndrome patients – A systematic review and meta-analysis

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TITLE PAGE

Prehospital risk assessment in suspected Acute Coronary Syndrome patients – A systematic review and meta-analysis

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ABSTRACT (259 words)

OBJECTIVE

To review, inventory and compare available diagnostic tools and investigate which tool has the best performance for pre-hospital risk assessment in suspected ACS patients.

METHODS

Systematic review and meta-analysis. Prospective studies with patients, suspected of NSTE-ACS, presenting pre-hospital in primary care or by Emergency Medical Services were included.

MAIN OUTCOME AND MEASURES

Sensitivity, specificity, and, likelihood ratio of findings for risk stratification in Acute Coronary Syndrome.

RESULTS

In total 15 prospective studies were included. No specific variables related to symptoms, physical examination or risk factors were useful in risk stratification for ACS diagnosis. The most useful electrocardiographic finding was ST-segment depression (LR+ 3.85 [95% CI, 2.58 to 5.76]). Point-of-care Troponin was found to be a strong predictor for ACS in primary care (LR+ 14.16 [95% CI, 4.28 to 46.90] and Emergency Medical Service setting (LR+ 6.16 [95% CI, 5.02 to 7.57]). Combined risk scores were the best for risk assessment in an ACS. From the combined risk scores that can be used immediately in a prehospital setting, the PreHEART score was most useful for risk stratification in patients with ACS (LR+ 8.19 [95% CI, 5.47 to 12.26]) and for identifying patients without ACS (LR- 0.05 [95% CI, 0.02 to 0.15]).

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CONCLUSION

In the prehospital setting several diagnostic tools have been reported which could improve risk stratification, triage and, early treatment in suspected ACS patients. On-site assessment of Troponin and combined risk scores derived from the HEART score are strong predictors. These results support further studies to investigate the impact of these new tools on logistics and clinical outcome.

KEYWORDS

NSTE-ACS, undifferentiated chest pain, prehospital, early triage, risk stratification, diagnostic tools, combined risk scores

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Key questions

What is already known about this subject?

The suspicion of an Acute Coronary Syndrome (ACS) is one of the main reasons for consultation of emergency medical care. Currently, only history and electrocardiogram are used in the prehospital setting for risk stratification.

What does this study add?

This systematic review and meta-analysis that included 15 prospective studies, shows that several new pre-hospital diagnostic tools for early risk-stratification are available. On-site assessment of point-of-care biomarkers and combined risk scores derived from the HEART score showed to be strong predictors for patients with a NSTE-ACS and to identify patients without a NSTE-ACS.

How might this impact on clinical practice?

Improvement of early risk assessment could improve triage, logistics and facilitates early treatment in ACS patients. These results support further studies to investigate the impact of these new tools on logistics and clinical outcome.

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INTRODUCTION

The suspicion of an Acute Coronary Syndrome (ACS) is one of the main reasons for consultation of emergency medical care.[1-4] Depending on national and regional protocols, patients who are suspected of having an ACS are first seen in primary care by a general practitioner (GP) or by the Emergency Medical Services (EMS). The decision to admit the suspected ACS patient directly to a tertiary intervention center, or to transport the patient to a local hospital for observation results from varying risk stratification methods. In general, this prehospital risk assessment and triage is mainly based on the patient's medical history and physical examination. None or only a limited number of diagnostic tools are used in this process. Often, only an ECG is made by the EMS to rule out a ST-segment elevation myocardial infarction (STEMI). Consequently, no adequate pre-hospital risk assessment and triage is performed in those without ST-segment elevation on their ECG (NSTE-ACS) and are transferred to a local hospital without onsite interventional options for further diagnostic evaluation.[2]

In the recent years, in-hospital diagnostic algorithm for suspected NSTE-ACS have been developed and improved. The implementation of combined risk scores such as the HEART and TIMI scores and new sensitive biomarker assays have substantially improved risk stratification, reduced the delay to diagnosis, shortened stays in the emergency departments and lowered health care costs.[5,6] Further improvement of care and cost reduction could be possible when these diagnostic algorithms would be applicable in the pre-hospital phase. The recent development of point-of-care (POC) analyzers for on-site biomarker assessment have accelerated these possibilities of pre-hospital risk assessment in suspected NSTE-ACS patients. Improving early risk stratification by GP or ambulance paramedics is important for several medical and economic reasons. Most importantly, due to the absence of reliable pre-hospital diagnostic tools, an NSTE-ACS is not always recognized, leading to a missed diagnosis in around 2-5% of consultations.[7] Additionally, the latest European guidelines recommend

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that high-risk NSTE-ACS patients receive an early invasive strategy within 24 hours.[8] It is well known that this is associated with shorter ischemic times and subsequent improved clinical outcomes.[9-11]

On the other hand, the use of additional diagnostic tools could help to definitively rule out an NSTE-ACS in the pre-hospital setting in chest pain patients with a low suspicion for NSTE-ACS. This is important as only 11-21% of these patients presenting to an emergency department are finally diagnosed with an NSTE-ACS, in the remaining majority generally no life-threatening condition is found.[12-14] Early risk assessment in these patients with low suspicion could significantly reduce the need of unnecessary diagnostics, referrals to hospitals, and therefore reduce health care cost.

Recently, several studies have been performed to investigate several known and new diagnostic tools for pre-hospital risk assessment in suspected ACS patients. We performed a systematic review and meta-analysis of available prospective data to summarize and compare current available and new diagnostic tools for pre-hospital risk assessment in suspected ACS patients, in particular those without ST-segment elevation on their ECG (NSTE-ACS).

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METHODS

Study Selection and Data Extraction

This meta-analysis was published for registration in PROSPERO prior to starting (CRD York, CRD42021254122).

We performed searches in MEDLINE and EMBASE for published articles. After identifying articles, we reviewed references from appropriate articles to identify additional references for this systematic review. Full search is shown in eAppendix 1.

Titles and abstract for all articles were screened by the primary author (J.D.) and decisions were checked by a second reviewer (P.J.V.). If the author identified the article as potentially suitable for inclusion, the full text of the article was reviewed in detail by two authors (J.D. and P.J.V.) and selected whether the inclusion criteria were met. If data were sufficient to generate a 2 x 2 table, the data was independently extracted, and the methodological quality and eligibility were determined. Bias was evaluated by the QUADAS Criteria.[15]

We included studies that met the following inclusion criteria: (1) patients suspected for NSTE-ACS; (2) prospective study; (3) original data ; (4) presenting pre-hospital (primary care, EMS); (5) prehospital risk assessment or triage; (6) outcome data on in-hospital ACS or MACE within 6 weeks after presentation; (7) article published in English, French, or German.

As the focus of this review was to investigate which patient characteristics and diagnostic tools were used for early risk assessment in an undifferentiated pre-hospital population suspected for ACS, we excluded studies enrolling a specific subpopulation from the general ACS population, studies with less than 100 patients, and studies enrolling only STEMI patients. Further, we excluded studies from which the data was incomplete, or the endpoints were not focused on diagnostic performance. Studies before 1995, the pre-troponin era, were excluded as well.

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End Points and Definitions

The primary end point was the final hospital discharge diagnosis of NSTE-ACS by the treating physician or Major Adverse Cardiac Event (MACE) within 6 weeks. NSTE-ACS refers to non-ST-elevation myocardial infarction (NSTEMI) or unstable angina pectoris (UAP). MACE was defined in the studies and included at least myocardial infarction.

Data and statistical analysis

For included studies individual data of sample size, number of true positive, true negative, false positive and false negative were extracted. When possible, we excluded the STEMI patients from the data extraction. We calculated sensitivity, specificity, likelihood ratios (LRs) and diagnostics odds ratio (dOR) from the extracted data for prediction of the primary end point. Extracted data included symptoms, physical examination, risk factors, ECG, biomarkers and Combined Risk Scores (CRS).

Since LRs is an appropriate way to measure and express diagnostic accuracy, our main focus was on this diagnostic measurement.[16,17] An LR around 1.0 has no proven diagnostic value. A positive likelihood ratio (LR+) between 1 and 2 as a minimal predictor for making ACS more likely, a LR+ of >2 as a weak predictor for ACS, a LR+ >5 as a moderate to strong predictor and a LR >10 as a very strong predictor. A negative likelihood ratio (LR-) between 0.5 and 1 as a minimal predictor to make ACS less likely, a LR- of <0.5 as a weak predictor, LR- < 0.2 as a moderate to strong predictor and a LR- <0.1 as a very strong predictor.[18]

Summary data for dichotomous findings are reported as weighted average when a finding was evaluated in 2 studies. When findings were reported in 3 or more studies the analyses were performed using R version 4.0.4 (R Foundation for Statistical Computing, Vienna, Austria). Sensitivity, specificity, and the diagnostic odds ratio analyses were performed using the 'meta' package as was heterogeneity. Likelihood ratios and bivariate analyses were performed using the 'mada' package.[19]

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Statistical heterogeneity was reported as the I^2 for the sensitivity and specificity when findings were evaluated in 3 studies or more. We considered values between 0% and 30% as low heterogeneity, between 30% and 60% as moderate heterogeneity and 60% or more as considerable heterogeneity.[20]

Patient and public involvement

Patient and public were not involved.

Statement of Ethics approval

It was not necessary to engage an ethics committee for this study-level systematic review and meta-analysis.

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RESULTS

Of the 1817 unique articles a total of 15 prospective studies met our inclusion criteria and were included in this systematic review **(Figure 1)**.[21-35] These studies reflected a total of 26083 patients. Five studies were performed in setting of primary care by the GP and 10 in the setting of the EMS. Rates of final NSTE-ACS diagnosis ranged in primary care from 5% to 22% (median 11% [interquartile range (IQR), 6%-20%]. In the EMS setting the rates of final ACS diagnosis ranged from 8% to 57% (median 17% [IQR, 12%-20%].

Data on 92 variables related to symptoms, physical examination, risk factors, ECG, biomarkers, and risk scores were extracted from the articles. The total number of studies providing data, the total number of patients, sensitivity, specificity, LR+, LR- and dOR are listed in tables 1-6. Risk of bias for each included study was scored with the QUADAS tool (**eAppendix 2)**.

Symptoms

Symptoms were not predictive for ACS in primary care or on the EMS setting **(eTable 1)**. The absence of having chest pain in the ambulance was a minimal predictor for making ACS less likely (LR- 0.61 [0.58-0.63]). Duration of symptoms in primary care until presentation were not appropriate for risk assessment in ACS.

Physical examination

Pain not reproducible by palpation (LR+ 1.10 [1.02-1.20]) and pallor (LR+ 1.17 [1.00-1.38]) were minimally predictive for making ACS more likely **(eTable 2)**. Reproducing the pain by palpation is a weak predictor for making ACS less likely (LR- 0.47 [0.43-0.51]) in EMS setting.

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Cardiovascular risk factors.

Of all the cardiovascular risk factors in EMS setting, history of prior coronary disease (LR+ 2.07 [1.58-2.71]) was the best predictor to make ACS more likely **(eTable 3)**. Male gender (LR+ 1.28 [1.12-1.45], history of angina pectoris (AP) (LR+ 1.27 [1.08-1.49]) or previous AMI (LR+ 1.83 [1.21-2.80]), CABG (LR+ 1.60 [1.14-2.24] or PCI (LR+ 1.49 [1.24-1.78]) were minimal predictors for ACS. For identifying patients less likely to have an ACS, no risk factor when absent conferred an LR- of 0.5 or lower. In primary care male gender (LR+ 1.61 [1.34-1.93]) was the only risk factor for making ACS more likely **(eTable 4)**.

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ECG

A total of 3 prospective studies investigated the value of ECG, 1 in primary care and 2 in EMS. [24,25,34] In all 3 studies the ECG was categorized in ischemic ECG versus abnormal and/or non-ischemic ECG. In only one study the exact definition of an ischemic ECG was mentioned: ST-deviation ≥800 µV. [21] Interpreting an ECG as an ischemic ECG is predictive for ACS in both the EMS setting (LR+ 2.23 [1.11-5.30]) as in primary care (LR+ 3.4 [2.15-5.37]). Classifying the ECG as non-ischemic is a weak predictor for making ACS less likely in primary care (LR- 0.33 [0.17-0.67]) and a minimal predictor in the EMS setting (LR- 0.57 [0.33-0.95]) (**Table 1)**. In terms of individual characteristics of the ECG, a Q-wave is a very strong predictor for ACS (LR+ 24.62 [3.38-179.20]. ST-depression (LR+ 3.85 [2.58-5.76]) is a better predictor for ACS than T-wave inversion (LR + 2.56 [1.50-4.39]) **(Table 1).**

Table 1. Performance of the ECG for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Ischemic ECG[b] [21,22]	EMS	2	1959	58 (45-71)	-	74 (59-85)	-	2.23 (1.11-5.30)	0.57 (0.33-0.95)	4.03 (0.70-23.42)
Ischemic ECG[b] [34]	GP	1	243	74 (53-88)	-	78 (69-84)	-	3.4 (2.15-5.37)	0.33 (0.17-0.67)	10.20 (3.55-29.28)
Q-wave [22]	EMS	1	541	11 (8-15)	-	100 (98-100)	-	25.36 (3.49-184.47)	0.89 (0.12-6.50)	28.39 (3.90-209.01)

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ST-depression [22]	EMS	1	541	42 (36-48)	-	89 (84-93)	-	3.85 (2.58-5.76)	0.65 (0.43-0.97)	5.91 (3.68-9.49)
T-wave inversion [22]	EMS	1	541	18 (14-23)	-	93 (89-96)	-	2.56 (1.50-4.39)	0.88 (0.52-1.51)	2.91 (1.62-5.22)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

[b] Definition of ischemic ECG given in 1 study only; ST-deviation ≥800 µV

Biomarkers

A total of 9 prospective studies investigated the value of biomarkers, 4 in primary care and 5 in EMS.[21-26,32-34] All blood samples for biomarkers were drawn in the pre-hospital setting. Most samples were analyzed directly prehospital by POC analyzers, but in 1 study the blood sample was analyzed after arrival in the hospital.[24] In EMS setting, median time of onset of complaint to biomarker assessment was 80 minutes (IQR 59–135 minutes) and in primary care the median was 600 (IQR 180-600 minutes). Further details on the prehospital analyzed biomarkers are shown in **eTable 5.**

Of the analyzed biomarkers in primary care the Heart-type Fatty Acid Binding Protein (H-FABP) (LR+ 7.84 [3.39-20.76]) was moderate to strong for predicting ACS **(Table 2).** POC-Troponin (LR + 14.16 [4.28-46.90]) was a very strong predictor for ACS. Both were not useful to make an ACS less likely (LR- 0.68 [0.31-1.91] vs LR- 0.71 [0.49-1.02]) **(Table 3)**.

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In the EMS setting, POC-Troponin (LR+ 6.16 [5.02-7.57]) was a moderate to strong predictor for making ACS more likely that could be interpreted immediately. There was no difference between blood sampled by POC in the EMS and blood analyzed in-hospital.

Table 2. Performance of the biomarkers for risk assessment in non-ST-elevation Acute Coronary Syndrome

Assay	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
H-FABP [23,25]	GP	2	601	35 (12-52)	-	96 (90-99)	-	7.84 (3.39-20.76)	0.68 (0.31-1.90)	11.53 (3.76-31.91)
Hs-Troponin T [24]	GP	1	115	83 (36-100)	-	76 (67-84)	-	3.49 (2.40-5.09)	0.22 (0.15-0.32)	15.97 (1.79-142.88)
Myoglobin [22]	EMS	1	536	8 (5-12)	-	96 (93-98)	-	2.07 (0.98-4.38)	0.96 (0.45-2.02)	2.17 (0.99-4.74)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; H-FABP; Heart-type Fatty Acid Binding Protein; HS, high-sensitive

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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Table 3. Performance of the POC-Troponin for risk assessment in non-ST-elevation Acute Coronary Syndrome

	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
POC-Troponin [22,27,28,32,33]	EMS	5	21484	25 (14-40)	96%	96 (94-98)	85%	6.16 (5.02-7.57)	0.75 (0.62-0.92)	8.27 (7.42-9.21)
POC-Troponin [24,26]	GP	2	311	31 (12-59)	-	98 (95-99)	-	14.16 (4.28-46.90)	0.71 (0.49-1.02)	20.01 (4.56-87.88)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; POC, Point-of-Care

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

Combined Risk Scores

Combined risk scores were investigated in 8 studies, 2 in primary care[25, 34] and 6 in EMS[28-32, 35]. In the EMS setting the used combined risk scores were all derived from the well-known HEART score.[5] In total 5 different combined risk scores were evaluated in this systematic review **(Table 4)**. There were minimal differences between the risk scores. These were due to different positivity threshold, risk factors and Troponin analyzers. In 2 studies data and blood were drawn prehospital but analyzed after arrival in-hospital.

In the EMS setting, all the combined risk scores were strong predictors for ACS when classified in the high-risk group. Between the risk scores there was no statistically significant difference **(Table 4).**

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In the low-risk group, the modified HEART performed the best in excluding ACS (LR-0.00). Of the combined risk scores that can be interpreted directly pre-hospital, preHEART is a very strong predictor in making ACS less likely (LR- (0.05 [0.02-0.15].

In all primary care studies, the combined risk scores classified patients in 2 risk groups (low vs intermediate/high). [25, 34] In the EMS setting this was only the case with the HEAR score and was a weak predictor for making ACS less likely (LR- 0.40 [0.38-0.43]).[32] The Marburg Heart Score (MHS) showed a significant better performance in making ACS less likely when the threshold value was lowered from 2 to 1 (LR- 0.57 [0.47-0.68] vs LR- 0.35 [0.32-0.38]) **(Table 5)**.

Alternative decision tools used in primary care, were the CRS. Scoring a positive answer to the elements, resulted in a point that was calculated into a final score. This final score classified the patient into a low risk group or intermediate-high risk group. CRS 1 had the best performance in making ACS less likely (LR- 0.24 [0.21-0.28]) and outperformed the MHS.

Compared to GP decision-making without use of a combined risk score (LR- 0.32 [0.00-0.46]) there is no improvement that makes ACS less likely **(Table 6)**.

Making an ACS more likely, decision making by the GP to refer to the hospital, without using a POC-troponin (LR+ 2.37 [1.95-3.73]) and using POC-Troponin (LR+ 3.20 [2.14-4.80]) outperformed the combined risk scores.

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Table 4. Performance of the Combined Risk Scores in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome

Risk Level	Studies	Threshold	LR (95-CI)	Troponin analyzer
High				
Core-lab PMHP [28]	1	Positive Troponin	8.68 (5.04-14.93)	In-hospital laboratorium
preHEART [35]	1	7-10	8.19 (5.47-12.26)	POC
PMHP [28]	1	Positive Troponin	6.70 (3.01-14.91)	POC
Modified HEART [30]	1	7-10	4.96 (3.62-6.79)	In-hospital laboratorium
HEART [35]	1	7-10	4.84 (3.41-6.89)	POC
Intermediate				
Core-lab PMHP [28]	1	≥4	0.61 (0.35-1.04)	In-hospital laboratorium
preHEART [35]	1	4-6	1.22 (0.95-1.57)	POC

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PMHP [28]	1	≥4	1.12 (0.74-1.70)	POC
Modified HEART [30]	1	4-6	0.89 (0.66-1.18)	Lab
HEART [35]	1	4-6	0.69 (0.49-0.98)	POC
Low				
Core-lab PMHP [28, 29]	2	0-3	0.08 (0-0.17)	In-hospital laboratorium
preHEART [35]	1	0-3	0.05 (0.02-0.15)	POC
PMHP [28]	1	0-3	0.26 (0.11-0.59)	POC
Modified HEART [30]	1	0-3	0.00	In-hospital laboratorium
HEART [31, 35]	2	0-3	0.14 (0.13-0.16)	POC
Abbreviations: 95-CI, 95% confidence interval; LR, likelihood ratio; POC. Point-of-Care; HEART, History, ECG, Age, Risk Factors, Troponin; PMHP, prehospital modified HEART pathway				

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Table 5. Performance of the Combined Risk Scores in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

CRS	Studies	Threshold for intermediate-high risk group	LR+ (95-CI)	Threshold for low-risk group	LR- (95-CI)	Setting	Elements of CRS
CRS 1 [25]	1	2-6	1.82 (1.59-2.10)	0-1	0.24 (0.21-0.28)	GP	• ST-depression • ST-elevation • Dyspnoe • Feeling of pressure chest pain • Absent laterale chest pain left • POC H-FABP +
CRS 2 [25]	1	2-5	1.69 (1.47-1.96)	0-1	0.36 (0.31-0.42)	GP	• ST-depression • ST-elevation • Dyspnoe • Feeling of pressure chest pain • Absent lateral chest pain left
CRS 3 [25]	1	2-4	1.75 (1.49-2.06)	0-1	0.40 (0.34-0.46)	GP	• Dyspnoe

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							• Feeling of pressure chest pain • Absent lateral chest pain left • POC H-FABP +
CRS 4 [25]	1	2-3	1.55 (1.30-1.85)	0-1	0.56 (0.47-0.67)	GP	• Dyspnoe • Feeling of pressure chest pain • Absent lateral chest pain left
MHS [34]	1	≥3	1.34 (1.11-1.61)	0-2	0.57 (0.47-0.68)	GP	
MHS [34]	1	≥2	1.12 (1.04-1.22)	0-1	0.35 (0.32-0.38)	GP	
GP probability assessment [34]	1	≥6	1.48 (1.29-1.70)	0-5	0.32 (0.28-0.37)	GP	
Combined MHS & GP probability assessment [34]	1	1-2	1.30 (1.19-1.42)	0	0	GP	
Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; CRS, Combined Risk Scores; HEAR, History, ECG, Age, Risk Factors; POC, Point-of-Care; H-FABP, Heart-type Fatty Acid Binding Protein; MHS, Marburg Heart Score; GP, general practitioner							

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Table 6. Performance of GP decision making for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
GP decision to refer [25 26]	2	371	79 (54-100)	-	67 (50-75)	-	2.37 (1.95-3.73)	0.32 (0.00-0.46)	7.50 (0-14.45)
GP decision to refer with POC-Troponin [26]	1	128	71 (29-96)	-	78 (69-85)	-	3.20 (2.14-4.80)	0.37 (0.25-0.55)	8.70 (1.60-47.40)
GP immediately suspected a serious condition [34]	1	243	58 (42-72)	-	53 (45-60)	-	1.22 (0.98-1.51)	0.80 (0.65-1.00)	1.51 (0.79-2.91)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; GP, General Practitioner; POC, Point-of-Care

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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Alternative diagnosis

In our systematic review 3 studies in the EMS setting described which alternative diagnoses were possible. From these, 2 studies noted the predictive value of POC-Troponin for other diseases[27,33], 1 study noted which other causes for chest pain could be classified as low-risk.[30] POC-Troponin is a moderate to strong predictor for other cardiac related diseases like myocarditis (LR+ 8.89 [8.29 – 9.54]), decompensated heart failure (LR+ 5.26 [1.88-6.03]), valve disorders/endocarditis (LR+ 3.29 [2.93-3.69]) and cardiomyopathy (LR+ 4.80 [4.36-5.29]). For non-cardiac diseases POC-Troponin is a weak to moderate predictor to make pulmonary embolism (LR+ 3.81 [1.38-4.46]), pneumonia (LR+ 2.41 [2.13-2.74]) and diseases of the genitourinary system more likely (LR+ 4.07 [2.66-4.51]). A normal POC-Troponin value makes a myocarditis less likely (LR- 0.40 [0.37-0.42]) but is not a strong predictor for other diagnoses. In the modified HEART, 403 patients were classified as low risk for ACS. In this group 2 serious non-cardiac diseased (pulmonary embolism, acute pancreatitis) were found.

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DISCUSSION

In this systematic review, we evaluated 15 prospective studies including 26.083 patients to identify patient characteristics and diagnostic tools for pre-hospital risk assessment in suspected NSTE-ACS patients. The main finding is that in the prehospital setting several new diagnostic tools are available which could improve early triage and risk assessment in suspected NSTE-ACS patients, in particular on-site assessment of biomarkers and combined risk scores derived from the HEART score were very strong for risk assessment in patients with NSTE-ACS.

Symptoms, physical examination, and risk factors

In accordance with previous analyses, in the present study no specific variables related to symptoms and physical examination were relevant predictors for an ACS diagnosis. The predictive value of some cardiovascular risk factors in the pre-hospital setting were as a single variable weak at best. A possible explanation is that in the pre-hospital setting less objective information is present on previous history of a patient. This was indeed corroborated in a recent analysis comparing pre-hospital and in-hospital calculated HEART scores.[29] In this study disagreement between hospital and pre-hospital HEART score risk classifications was found in approximately 25% of patients, mainly by different scoring of history and cardiovascular risk factors.

ECG

In patients with suspected ACS, the resting 12-lead ECG is the first-line diagnostic tool. Our results indicate that in the pre-hospital setting only a minority of the studies provided detailed description of the ECG criteria and used definitions. Moreover, the ECG was only a moderate predictor for NSTE-ACS. We expect that a more detailed analysis of ECGs, training of EMS personnel or GPs could improve the performance of ECG in the prehospital setting.

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Biomarkers

The advent of POC analyzers facilitates the use of established biomarkers in the pre-hospital setting. Current guidelines state that measurement of the biomarker Troponin is mandatory in all suspected ACS patients.[36] Although the included studies used normal Troponin I and T assays, the first high-sensitivity assays also became available recently for POC analyzers**.** This can increase diagnostic accuracy and reduce the time to diagnosis.[37] Depending on the type of assay used, assessment of this biomarker is already reliable 3-6 hours after onset of symptoms. The duration of symptoms could differ between different settings and populations. We found that duration of symptoms often is much longer in the primary care setting compared to the EMS setting. This explains why the negative predictive value of POC troponin is much higher in the primary care setting than in EMS setting. Consequently, symptom duration could indeed be too short to rule out an ACS based on a single troponin assessment in the EMS setting. However, the use of POC troponin in a combined risk score could offer a solution to lower the risk of false negatives. This is supported by the results from our analyses which clearly show that the combined risk scores are superior to a single troponin for risk assessment in a patient suspected for NSTE-ACS.

Two primary care studies[23 25] examined the diagnostic value of H-FABP, a small protein that is common in cardiac muscle and is released within 2 hours after the onset of ischemia.[38] Although there are no prehospital studies that compare Troponin with H-FABP, several in-hospital studies have compared these biomarkers. Despite of the early release, these studies showed that the predictive value of the H-FABP is moderate when assessed within the time frame of 6 hours after onset of symptoms. Compared to POC-Troponin, stand-alone H-FABP showed inferior diagnostic performance.[39,40,41]

Combined risk scores

International guidelines advise the use of combined risk scores at emergency departments.[36] Several scores are available that, generally combine elements of history, cardiovascular risk factors, ECG, and

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the biomarker Troponin. The advent of POC analyzers makes it possible to use these risk scores in the prehospital setting. We found mainly risk scores derived from the well-known HEART score, which were validated for prehospital usage. The results tend to be very promising; for identifying high-risk patients for NSTE-ACS, but especially for excluding ACS for low-risk patients. Compared to a previous systematic review by Fanaroff et al.[42] in 2015, in-hospital use of the original HEART score shows higher predictive value for high-risk patients in diagnosing ACS, and a similar strong negative predictive value is found for patients classified as low-risk. The better predictive value for the high-risk group in the original, in-hospital HEART score can be explained by the presence of more objective information and the longer time between the onset of symptoms in relation to blood analyses. In this patient population there exists a considerable probability that the in-hospital troponin value will be positive more often compared to the prehospital setting as explained before.

In the present analysis, the performance of combined risk scores in the primary care setting were lower as compared with the EMS setting. This is due the fact that different scores were used, and in none a POC troponin. In addition, important differences exist in patient selection and prevalence of ACS, which also could have impacted the results. Finally, the performance of diagnostic tools (in particular ECG) and combined risk scores depends on the experience and support of each individual physician. In primary care there are large difference between regions and countries to what extent additional diagnostic tools are used or necessary. In particular in remote areas with no nearby hospitals, the implementation of combined risk scores could improve acute medical care for suspect ACS patients.

When evaluating a patient with suspected ACS, life-threatening diagnoses such as aortic dissection and pulmonary embolism should be part of the clinical evaluation and differential diagnosis. These alternative diagnoses are often associated with increased levels of biomarkers or ECG changes as well. So, the question is if the prehospital tested biomarkers and combined risk score are sufficient to

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also rule out chest pain due to aortic dissection or pulmonary embolism. Our meta-analysis included 3 studies that reported life-threatening diagnoses in the different risk groups. In the group of patients classified as low-risk for ACS, one pulmonary embolism and one acute pancreatitis occurred. No information regarding aortic dissections was available. These data indicate that although a combined risk score is a very strong predictor for excluding ACS, the possibility for a severe alternative diagnosis should not be ignored.

Further perspective and upcoming studies

Before using these new diagnostic tools in routine practice, more data on safety, efficacy and, performance are necessary. More data are needed when these combined risk scores are used to determine a different triage and/or treatment strategy. Some up-coming studies that will examine whether this early risk stratification and associated referral decisions based on combined risk score are feasible, are the preHEART3 study (Trialregister.nl, NL7866), PRESTO study[43], ARTICA trial[44], FamouS Triage 3 study[45] and POPular HEART (Clinicaltrials.gov, NCT04851418).

Study limitations

Although we included only prospective studies to improve the overall study quality, this analysis suffers from several limitations inherent to this study design and patient population.

An important limitation of the study was verification bias. There was a variable definition for ACS between the studies, for example not all studies included unstable angina pectoris and in some studies some patients were included who developed a STEMI. Although this reflects daily practice, it could have impacted the results.

In the analysis of some variables, heterogeneity was considerable in our meta-analysis (e.g., cardiovascular risk factors and POC-Troponin measurement in the EMS setting). In POC-Troponin,

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heterogeneity is the result of variation in thresholds between studies. Although this is expected in systematic reviews of diagnostic test accuracy, it is possible this influences our main findings.[46] However, given the small contribution we do not think that the heterogeneity in our meta-analysis alternates the influence of stand-alone risk factors in early risk stratification for ACS.

Additionally, many of the diagnostic tools were examined in a limited number of studies and study populations. Given the large differences in patient presentation and prevalence of ACS, the performance of the different diagnostic tools could differ when used in a clinical setting or population with different prevalence of ACS.

Finally, not for all diagnostic tools information on false negative results and alternative life-threatening diagnoses were given. Before changing triage and consequent treatment based on these tools, more prospective data are necessary to assess the safety in these rare but important selection of patients.

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Conclusion

In the pre-hospital setting several new diagnostic tools are available which could improve risk stratification, triage and, early treatment in suspected ACS patients. On-site assessment of biomarkers and combined risk scores derived from the HEART score showed to be strong predictors for diagnosing patients with a NSTE-ACS and to identify patients without a NSTE-ACS. These promising results underline the importance of new prospective studies in this field.

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Contributorship Statement:

J.D: planning, conducting, data collection, reporting, and check, statistical analysis, article writing, submission

J.Z.: data collection and reporting, statistical analysis, article writing

A.K.: article writing

G.S.: article writing

P.H.: article writing

P.T.: article writing

L.D.: article writing

M.V.: planning, conducting, date collection, reporting, and check, statistical check, article writing

P.V.: planning, conducting, data collection and reporting, article writing, submission

The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted

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Conflict of interest statement:

None.

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Legends

Figure 1 is the graphical abstract. It shows the indicative positive and negative likelihood ratio for each element and how it compares to its predictive value.

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Demandt et al. Prehospital risk assessment in ACS.

References

1. Hoorweg BB, Willemsen RT, Cleef LE, et al. Frequency of chest pain in primary care, diagnostic tests performed and final diagnoses. *Heart* 2017;103(21):1727-32. doi: 10.1136/heartjnl-2016-310905 [published Online First: 2017/06/22]
2. Pedersen CK, Stengaard C, Friesgaard K, et al. Chest pain in the ambulance; prevalence, causes and outcome - a retrospective cohort study. *Scand J Trauma Resusc Emerg Med* 2019;27(1):84. doi: 10.1186/s13049-019-0659-6 [published Online First: 2019/08/30]
3. Thang ND, Karlson BW, Bergman B, et al. Characteristics of and outcome for patients with chest pain in relation to transport by the emergency medical services in a 20-year perspective. *Am J Emerg Med* 2012;30(9):1788-95. doi: 10.1016/j.ajem.2012.02.014 [published Online First: 2012/05/29]
4. Wibring K, Lingman M, Herlitz J, et al. Prehospital stratification in acute chest pain patient into high risk and low risk by emergency medical service: a prospective cohort study. *BMJ Open* 2021;11(4):e044938. doi: 10.1136/bmjopen-2020-044938 [published Online First: 2021/04/17]
5. Backus BE, Six AJ, Kelder JC, et al. A prospective validation of the HEART score for chest pain patients at the emergency department. *Int J Cardiol* 2013;168(3):2153-8. doi: 10.1016/j.ijcard.2013.01.255
6. Antman EM, Cohen M, Bernink PJ, et al. The TIMI risk score for unstable angina/non-ST elevation MI: A method for prognostication and therapeutic decision making. *JAMA* 2000;284(7):835-42. doi: 10.1001/jama.284.7.835
7. Harskamp R, van Peet P, Bont J, et al. The conundrum of acute chest pain in general practice: a nationwide survey in The Netherlands. *BJGP Open* 2018;2(4):bjgpopen18X101619. doi: 10.3399/bjgpopen18X101619
8. Collet JP, Thiele H, Barbato E, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Rev Esp Cardiol (Engl Ed)* 2021;74(6):544. doi: 10.1016/j.rec.2021.05.002 [published Online First: 2021/05/23]
9. Bainey KR, Gafni A, Rao-Melacini P, et al. The cost implications of an early versus delayed invasive strategy in Acute Coronary Syndromes: the TIMACS study. *J Med Econ* 2014;17(6):415-22. doi: 10.3111/13696998.2014.911184 [published Online First: 2014/04/08]
10. Nieuwets A, Poldervaart JM, Reitsma JB, et al. Medical consumption compared for TIMI and HEART score in chest pain patients at the emergency department: a retrospective cost analysis. *BMJ Open* 2016;6(6):e010694. doi: 10.1136/bmjopen-2015-010694 [published Online First: 2016/06/18]
11. de Koning E, Biersteker TE, Beeres S, et al. Prehospital triage of patients with acute cardiac complaints: study protocol of HART-c, a multicentre prospective study. *BMJ Open* 2021;11(2):e041553. doi: 10.1136/bmjopen-2020-041553 [published Online First: 2021/02/14]
12. Goodacre S, Cross E, Arnold J, et al. The health care burden of acute chest pain. *Heart* 2005;91(2):229-30. doi: 10.1136/hrt.2003.027599
13. Kienbacher CL, Fuhrmann V, van Tulder R, et al. Impact of more conservative European Society of Cardiology guidelines on the management of patients with acute chest pain. *Int J Clin Pract* 2021;75(6):e14133. doi: 10.1111/ijcp.14133 [published Online First: 2021/03/09]
14. Boubaker H, Grissa MH, Beltaief K, et al. A new score for the diagnosis of acute coronary syndrome in acute chest pain with non-diagnostic ECG and normal troponin. *Emerg Med J* 2015;32(10):764-8. doi: 10.1136/emermed-2013-203151 [published Online First: 2015/01/07]
15. Whiting PF, Weswood ME, Rutjes AW, et al. Evaluation of QUADAS, a tool for the quality assessment of diagnostic accuracy studies. *BMC Med Res Methodol* 2006;6:9. doi: 10.1186/1471-2288-6-9 [published Online First: 2006/03/08]

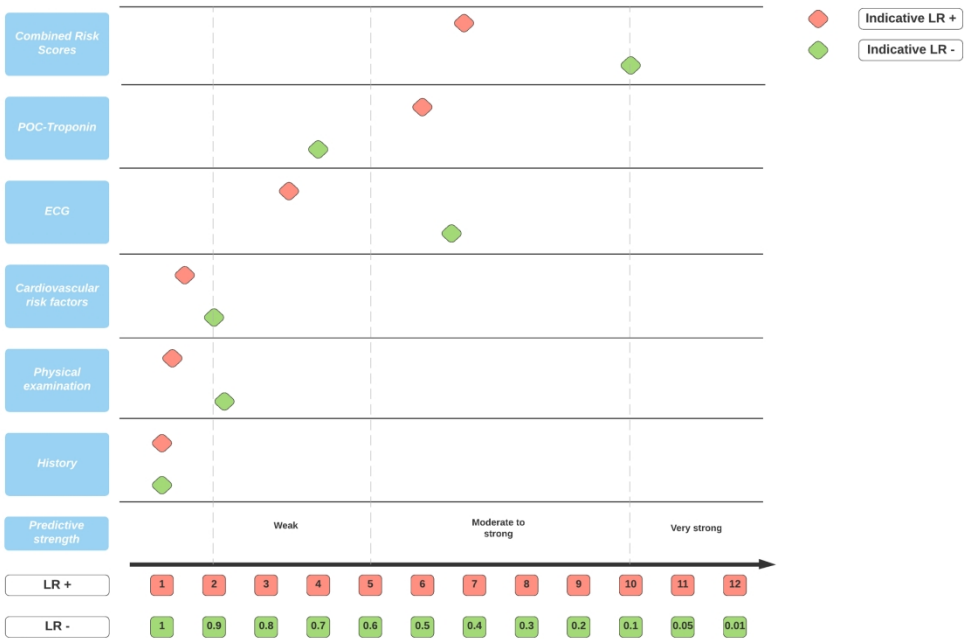
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. McGee S. Simplifying likelihood ratios. *J Gen Intern Med* 2002;17(8):646-9. doi: 10.1046/j.1525-1497.2002.10750.x
17. Honest H, Khan KS. Reporting of measures of accuracy in systematic reviews of diagnostic literature. *BMC Health Serv Res* 2002;2:4. doi: 10.1186/1472-6963-2-4
18. Grimes DA, Schulz KF. Refining clinical diagnosis with likelihood ratios. *Lancet* 2005;365(9469):1500-5. doi: 10.1016/S0140-6736(05)66422-7
19. Shim SR, Kim SJ, Lee J. Diagnostic test accuracy: application and practice using R software. *Epidemiol Health* 2019;41:e2019007. doi: 10.4178/epih.e2019007
20. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21(11):1539-58. doi: 10.1002/sim.1186
21. Anroedh SS, Kardys I, Akkerhuis KM, et al. e-Transmission of ECGs for expert consultation results in improved triage and treatment of patients with acute ischaemic chest pain by ambulance paramedics. *Neth Heart J* 2018;26(11):562-71. doi: 10.1007/s12471-018-1187-0 [published Online First: 2018/10/26]
22. Svensson L, Isaksson L, Axelsson C, et al. Predictors of myocardial damage prior to hospital admission among patients with acute chest pain or other symptoms raising a suspicion of acute coronary syndrome. *Coron Artery Dis* 2003;14(3):225-31. doi: 10.1097/01.mca.0000063503.13456.0d [published Online First: 2003/04/19]
23. Bruins Slot MH, Rutten FH, van der Heijden GJ, et al. Diagnostic value of a heart-type fatty acid-binding protein (H-FABP) bedside test in suspected acute coronary syndrome in primary care. *Int J Cardiol* 2013;168(2):1485-9. doi: 10.1016/j.ijcard.2012.12.050 [published Online First: 2013/01/22]
24. Andersson PO, Karlsson JE, Landberg E, et al. Consequences of high-sensitivity troponin T testing applied in a primary care population with chest pain compared with a commercially available point-of-care troponin T analysis: an observational prospective study. *BMC Res Notes* 2015;8:210. doi: 10.1186/s13104-015-1174-0 [published Online First: 2015/06/04]
25. Willemsen RT, Winkens B, Kietselaer BL, et al. Evaluating possible acute coronary syndrome in primary care: the value of signs, symptoms, and plasma heart-type fatty acid-binding protein (H-FABP). A diagnostic study. *BJGP Open* 2019;3(3) doi: 10.3399/bjgpopen19X101652 [published Online First: 2019/10/04]
26. Nilsson S, Andersson PO, Borgquist L, et al. Point-of-Care Troponin T Testing in the Management of Patients with Chest Pain in the Swedish Primary Care. *Int J Family Med* 2013;2013:532093. doi: 10.1155/2013/532093 [published Online First: 2013/02/01]
27. Stengaard C, Sorensen JT, Ladefoged SA, et al. Quantitative point-of-care troponin T measurement for diagnosis and prognosis in patients with a suspected acute myocardial infarction. *Am J Cardiol* 2013;112(9):1361-6. doi: 10.1016/j.amjcard.2013.06.026 [published Online First: 2013/08/21]
28. Stopyra JP, Snavely AC, Smith LM, et al. Prehospital use of a modified HEART Pathway and point-of-care troponin to predict cardiovascular events. *PLoS One* 2020;15(10):e0239460. doi: 10.1371/journal.pone.0239460 [published Online First: 2020/10/08]
29. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Accuracy of pre-hospital HEART score risk classification using point of care versus high sensitive troponin in suspected NSTE-ACS. *Am J Emerg Med* 2020;38(8):1616-20. doi: 10.1016/j.ajem.2019.158448 [published Online First: 2019/11/09]
30. Ishak M, Ali D, Fokkert MJ, et al. Fast assessment and management of chest pain patients without ST-elevation in the pre-hospital gateway (FamouS Triage): ruling out a myocardial infarction at home with the modified HEART score. *Eur Heart J Acute Cardiovasc Care* 2018;7(2):102-10. doi: 10.1177/2048872616687116 [published Online First: 2017/01/14]
31. van Dongen DN, Tolsma RT, Fokkert MJ, et al. Pre-hospital risk assessment in suspected non-ST-elevation acute coronary syndrome: A prospective observational study. *Eur Heart J Acute Cardiovasc Care* 2020;9(1_suppl):5-12. doi: 10.1177/2048872618813846 [published Online First: 2018/11/24]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Value of Prehospital Troponin Assessment in Suspected Non-ST-Elevation Acute Coronary Syndrome. *Am J Cardiol* 2018;122(10):1610-16. doi: 10.1016/j.amjcard.2018.07.037 [published Online First: 2018/09/29]
33. Rasmussen MB, Stengaard C, Sorensen JT, et al. Predictive value of routine point-of-care cardiac troponin T measurement for prehospital diagnosis and risk-stratification in patients with suspected acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;8(4):299-308. doi: 10.1177/2048872617745893 [published Online First: 2017/12/05]
34. Schols AMR, Willemsen RTA, Bonten TN, et al. A Nationwide Flash-Mob Study for Suspected Acute Coronary Syndrome. *Ann Fam Med* 2019;17(4):296-303. doi: 10.1370/afm.2401 [published Online First: 2019/07/10]
35. Sagel D, Vlaar PJ, van Roosmalen R, et al. Prehospital risk stratification in patients with chest pain. *Emerg Med J* 2021 doi: 10.1136/emermed-2020-210212 [published Online First: 2021/08/11]
36. Collet JP, Thiele H, Barbato E, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Eur Heart J* 2021;42(14):1289-367. doi: 10.1093/eurheartj/ehaa575 [published Online First: 2020/08/30]
37. Boeddinghaus J, Nestelberger T, Koechlin L, et al. Early Diagnosis of Myocardial Infarction With Point-of-Care High-Sensitivity Cardiac Troponin I. *J Am Coll Cardiol* 2020;75(10):1111-24. doi: 10.1016/j.jacc.2019.12.065
38. Dekker MS, Mosterd A, van 't Hof AW, et al. Novel biochemical markers in suspected acute coronary syndrome: systematic review and critical appraisal. *Heart* 2010;96(13):1001-10. doi: 10.1136/hrt.2009.189886
39. Lippi G, Mattiuzzi C, Cervellin G. Critical review and meta-analysis on the combination of heart-type fatty acid binding protein (H-FABP) and troponin for early diagnosis of acute myocardial infarction. *Clin Biochem* 2013;46(1-2):26-30. doi: 10.1016/j.clinbiochem.2012.10.016
40. Bivona G, Agnello L, Bellia C, et al. Diagnostic and prognostic value of H-FABP in acute coronary syndrome: Still evidence to bring. *Clin Biochem* 2018;58:1-4. doi: 10.1016/j.clinbiochem.2018.04.021
41. Bruins Slot MH, van der Heijden GJ, Stelpstra SD, et al. Point-of-care tests in suspected acute myocardial infarction: a systematic review. *Int J Cardiol* 2013;168(6):5355-62. doi: 10.1016/j.ijcard.2013.08.002 [published Online First: 2013/09/05]
42. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does This Patient With Chest Pain Have Acute Coronary Syndrome?: The Rational Clinical Examination Systematic Review. *JAMA* 2015;314(18):1955-65. doi: 10.1001/jama.2015.12735
43. Alghamdi A, Cook E, Carlton E, et al. PRe-hospital Evaluation of Sensitive TrOponin (PRESTO) Study: multicentre prospective diagnostic accuracy study protocol. *BMJ Open* 2019;9(10):e032834. doi: 10.1136/bmjopen-2019-032834
44. Aarts GWA, Camaro C, van Geuns RJ, et al. Acute rule-out of non-ST-segment elevation acute coronary syndrome in the (pre)hospital setting by HEART score assessment and a single point-of-care troponin: rationale and design of the ARTICA randomised trial. *BMJ Open* 2020;10(2):e034403. doi: 10.1136/bmjopen-2019-034403
45. van Dongen DN, Tolsma RT, Fokkert MJ, et al. Referral decisions based on a prehospital HEART score in suspected non-ST-elevation acute coronary syndrome: design of the FamouS Triage 3 study. *Future Cardiol* 2020;16(4):217-26. doi: 10.2217/fca-2019-0030
46. Macaskill P, Gatsonis C, Deeks JJ, et al. Chapter 10: Analysing and Presenting Results. In: Deeks JJ, Bossuyt PM, Gatsonis C, eds. Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy Version 10. http://srdta.cochrane.org/: The Cochrane Collaboration 2010.

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228x155mm (300 x 300 DPI)

Supplemental material

Supplement to: Demandt JPA, Zelis JM, Koks A, et al. Prehospital Risk Assessment In Suspected Acute Coronary Syndrome Patients – A Systematic Review And Meta-analysis

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eAppendix 1. Full search

eAppendix 2. Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool

eFigure 1. Flow diagram of study inclusion

eTable 1. Performance of symptoms for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 2. Performance of physical examination for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 3. Performance of risk factors in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 4. Performance of risk factors in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 5. Additional information about Point-of-Care (POC) analyzers

eTable 6. Study Characteristics

eTable 7. Results of individual studies of symptoms

eTable 8. Results of individual studies of risk factors

eTable 9. Results of individual studies of electrocardiogram findings

eTable 10. Results of individual studies of biomarkers

eTable 11. Results of individual studies of Combined Risk Scores

eTable 12. Results of individual studies of general practitioner decision making

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eAppendix 1. Full search

Search strategy in MEDLINE:
(ambulance OR pre-hospital OR prehospital OR triage OR paramedics OR ems) AND (ecg OR troponin OR "risk scores" OR POC) AND (acute coronary syndrome OR acs OR nstemi OR "chest pain" OR (acute coronary syndrome [MeSH Terms]) OR (chest pain[MeSH Terms])).

Specific for primary care studies, we performed a second search:
("general practitioner" OR "primary care" OR outpatient) AND (ecg OR troponin OR "risk scores" OR POC) AND (acute coronary syndrome OR acs OR nstemi OR "chest pain" OR (acute coronary syndrome [MeSH Terms]) OR (chest pain[MeSH Terms]))

For our EMBASE search, we added a limitation to exclude all MEDLINE journals and we replaced the MeSH terms with the appropriate Emtree terms.
((ambulance or pre-hospital or prehospital or triage or paramedics or ems).mp. or exp ambulance/) and (ecg or troponin or risk scores or POC).mp. and ((acute coronary syndrome or ACS or NSTEMI or chest pain).mp. or exp non st segment elevation acute coronary syndrome/ or exp acute coronary syndrome/).

In EMBASE we performed a second search as well for additional primary care studies.
(general practitioner OR primary care OR outpatient OR exp general practitioner/ OR primary health care/) and (ecg or troponin or risk scores or POC) and (acute coronary syndrome or ACS or NSTEMI or chest pain or exp non st segment elevation acute coronary syndrome/ or exp acute coronary syndrome/).

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eAppendix 2. Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool[1]

QUADAS tool: description

1. Was the spectrum of patients representative of the patients who will receive the test in practice?
   Patients presenting in a prehospital setting (EMS, GP) with suspected ACS = yes, retrospective=no

2. Is the reference standard likely to classify the target condition correctly?
   If reference standard in hospital adjudication of ACS, especially NSTEMI or unstable AP; if reference standard is Major Adverse Cardiac Events (MACE), including at least acute myocardial infarction (AMI) =yes

3. Is the time period between reference standard and index test short enough to be reasonably sure that the target condition did not change between the two tests?
   Adjudicated diagnosis at dismissal. MACE within 6 weeks=yes

4. Did the whole sample or a random selection of the sample, receive verification using a reference standard of diagnosis?

5. Did patients receive the same reference standard irrespective of the index test result?

6. Was the reference standard independent of the index test (i.e., the index test did not form part of the reference standard)?

7. Were the reference standard results interpreted without knowledge of the results of the index test? (Index test results blinded)

8. Were the index test results interpreted without knowledge of the results of the reference standard? (Reference standard results blinded)

9. Were the same clinical data available when test results were interpreted as would be available when the test is used in practice?
   When the test executor had as much info as in clinical practice=yes

10. Were uninterpretable/intermediate test results reported?
    Not reported, numbers are correct=yes

11. Were withdrawals from the study explained?
    Not reported, numbers are correct=yes

Did the study provide a clear definition of what was to be considered a 'positive' result?

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QUADAS tool: judgement

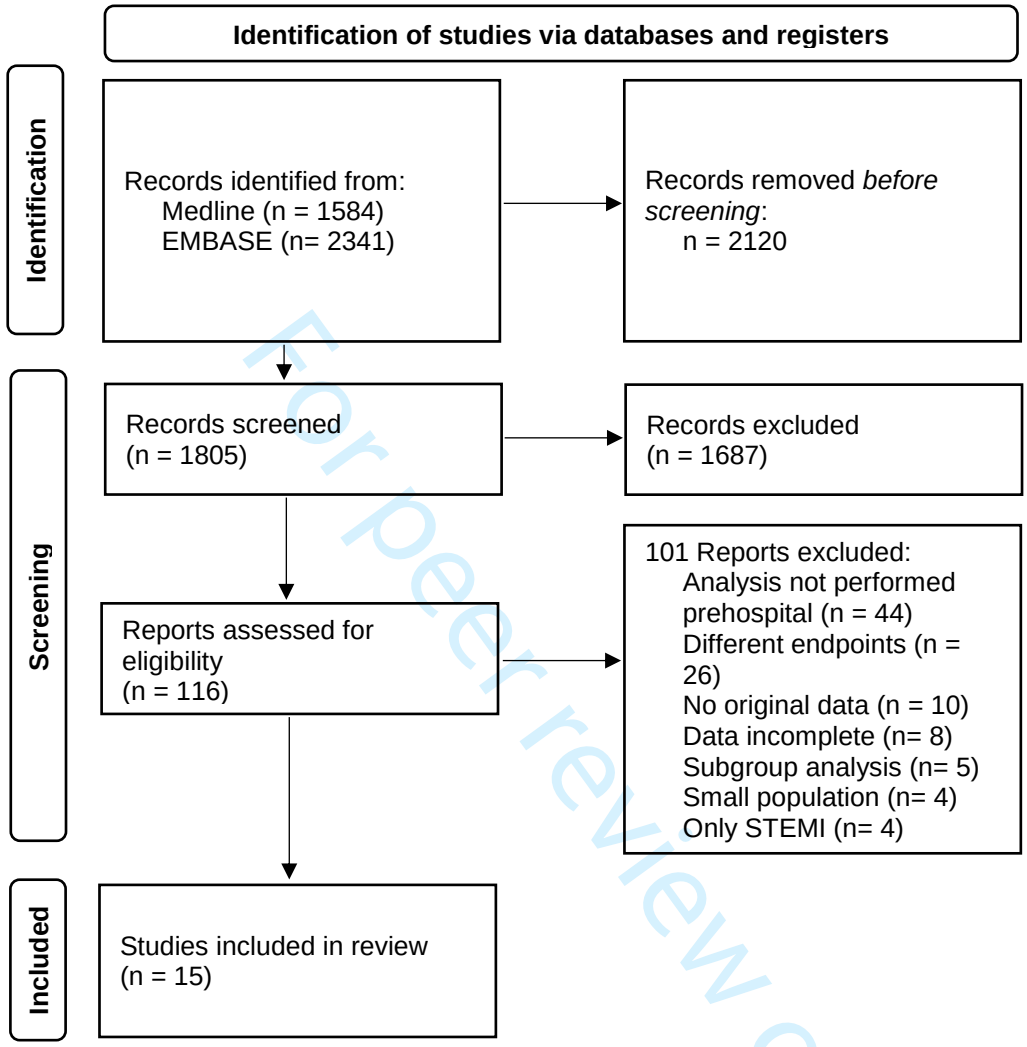
Study	1	2	3	4	5	6	7	8	9	10	11	A
Van Dongen 2018 (16)	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Van Dongen 2020 (17)	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Ishak 2018 (18)	Y	Y	Y	Y	Y	N	U	U	Y	N	N	Y
Van Dongen 2020 (19)	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Bruins Slot 2013 (20)	Y	Y	Y	N	Y	Y	Y	Y	Y	N	Y	Y
Andersson 2015 (21)	Y	Y	Y	Y	Y	N	Y	Y	Y	N	Y	Y
Schols 2019 (22)	Y	Y	Y	Y	U	Y	Y	Y	Y	Y	Y	U
Willemsen 2019 (23)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Nilsson 2013 (24)	Y	Y	Y	Y	Y	N	Y	Y	Y	N	Y	Y
Anroedh 2018 (25)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	-
Svensson 2003 (26)	Y	Y	Y	Y	Y	N	U	U	Y	N	N	Y
Stopyra 2020 (27)	Y	Y	Y	Y	Y	N	U	Y	Y	N	N	Y
Stengaard 2013 (28)	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y
Sagel 2021 (29)	Y	Y	Y	Y	Y	N	U	Y	Y	N	Y	Y
Rasmussen 2019 (29)	Y	Y	U	Y	Y	N	N	Y	Y	N	Y	Y

Y = yes; N = no

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eFigure 1. Flow diagram of study inclusion



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eTable 1.

eTable 1. Performance of symptoms for risk assessment in non-ST-elevation Acute Coronary Syndrome										
Symptom	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Chestpain[22]	EMS	1	538	96 (93-98)	-	7 (4-11)	-	1.03 (0.98-1.07)	0.61 (0.58-0.63)	1.70 (0.80-3.61)
Dyspnoe [22]	EMS	1	536	49 (43-55)	-	46 (39-53)	-	0.90 (0.76-1.07)	1.12 (0.94-1.32)	0.81 (0.57-1.14)
Nausea [22]	EMS	1	536	37 (32-43)	-	71 (65-77)	-	1.29 (1.00-1.67)	0.88 (0.68-1.14)	1.46 (1.01-2.11)
Chestpain [34]	GP	1	243	87 (73-95)	-	11 (7-16)	-	0,98 (0.90-1.05)	1.29 (1.11-1.29)	0.81 (0.31-2.14)
Worse with exertion [34]	GP	1	212	47 (30-65)	-	55 (46-63)	-	1.0 (0.79-1.38)	0.97 (0.73-1.28)	1.08 (0.52-2.23)
Chestpain feels like pressure [34]	GP	1	212	79 (64-91)	-	11 (7-17)	-	0.89 (0.81-0.98)	1.87 (1.70-2.05)	0.48 (0.19-1.19)
Patient assumes cardiac origin of pain [34]	GP	1	242	62 (47-76)	-	45 (38-52)	-	1.13 (0.94-1.37)	0.84 (0.69-1.01)	1.36 (0.70-2.64)
Symptoms duration <1h [34]	GP	1	243	87 (73-95)	-	11 (7-16)	-	0.98 (0.90-1.05)	1.2 (1.11-1.29)	0.81 (0.31-2.14)

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Symptoms duration (1-24h) [34]	GP	1	243	51 (36-66)	-	42 (35-49)	-	0.88 (0.72-1.08)	1.16 (0.95-1.43)	0.75 (0.39-1.44)
Symptoms duration (>24h)[34]	GP	1	243	42 (28 58)	-	68 (61-75)	-	1.33 (0.97-1.82)	0.85 (0.62-1.16)	1.57 (0.81-3.04)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 2

eTable 2. Performance of physical examination for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Pale [22]	EMS	1	536	60 (54-65)	-	49 (42-56)	-	1.17 (1.00-1.38)	0.82 (0.70-0.96)	1.43 (1.01-2.02)
Clammy [22]	EMS	1	536	37 (32-43)	-	68 (62-74)	-	1.16 (0.21-1.49)	0.92 (0.72-1.18)	1.26 (0.88-1.81)
Pain not reproducible by palpation [34]	GP	1	210	92 (79-98)	-	16 (11-23)	-	1.10 (1.02-1.20)	0.47 (0.43-0.51)	2.35 (0.68-8.16)
Cardiac murmur [25]	GP	1	250	4 (0-19)	-	91 (86-94)	-	0.41 (0.08-1.91)	1.06 (0.23-4.89)	0.39 (0.05-3.03)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 3

eTable 3. Performance of risk factors in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome										
Risk factor	Setting	Studies	Patients	Sensitivity % (95-CI)	I^2, %[a]	Specificity % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Prior coronary disease [28]	EMS	1	389	50 (38-62)	-	76 (71-80)	-	2.07 (1.58-2.71)	0.66 (0.50-0.86)	3.14 (1.86-5.31)
Previous PAV [28,32]	EMS	2	1095	8 (1-16)	-	96 (92-98)	-	1.93 (0.38-5.53)	0.96 (0.37-2.70)	2.01 (0.33-6.35)
Previous AMI [22,27,28,32]	EMS	4	2618	39 (31-48)	88%	79 (69-88)	96%	1.84 (1.21-2.80)	0.80 (0.71-0.91)	2.32 (1.34-4.00)
Previous CABG [27,28,32]	EMS	3	2080	10 (8-14)	0%	93 (90-95)	84%	1.60 (1.14-2.24)	0.96 (0.93-0.99)	1.67 (1.15-2.44)
Previous PCI [27,28,32]	EMS	3	2080	32 (28-36)	13%	80 (74-85)	86%	1.49 (1.24-1.78)	0.87 (0.81-0.93)	1.71 (1.35-2.17)
Previous TIA/Stroke [28,32]	EMS	2	1095	9 (4-19)	-	93 (86-96)	-	1.30 (0.45-3.21)	0.98 (0.52-2.00)	1.33 (0.37-3.7)
Current smoker [22,27,28,32]	EMS	4	2618	27 (23-33)	63%	79 (74-84)	90%	1.29 (0.78-2.14)	0.93 (0.80-1.07)	1.39 (0.72-2.68)
Male gender [22,27,28,32]	EMS	4	2618	67 (59-75)	85%	48 (44-52)	74%	1.28 (1.12-1.45)	0.69 (0.53-0.90)	1.86 (1.39-2.80)

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AP [22]	EMS	1	536	64 (58-69)	-	50 (43-56)	-	1.27 (1.08-1.49)	0.73 (0.62-0.85)	1.75 (1.24-2.48)
Hypercholesterolemia [27,28,32]	EMS	3	2080	60 (34-81)	97%	50 (22-79)	99%	1.23 (0.93-1.61)	0.83 (0.73-0.94)	1.47 (1.04-2.10)
DM [22,27,28,32]	EMS	4	2618	23 (16-31)	83%	81 (75-86)	90%	1.22 (0.95-1.57)	0.96 (0.89-1.03)	1.29 (0.93-1.80)
Heart failure [22,28]	EMS	2	933	22 (11-30)	-	82 (69-91)	-	1.21 (0.65-2.49)	0.95 (0.56-1.52)	1.27 (0.57-3.2)
Hypertension [22,27,28,32]	EMS	4	2618	56 (41-70)	94%	52 (37-66)	96%	1.14 (1.04-1.25)	0.88 (0.81-0.95)	1.33 (1.10-1.61)
Obesity [28, 32]	EMS	2	1095	29 (16-51)	-	70 (45-84)	-	0.99 (0.65-1.62)	1.00 (0.70-1.47)	0.99 (0.40-2.04)
Previous smoker [27]	EMS	1	985	30 (24-36)	-	72 (69-76)	-	1.08 (0.89-1.31)	0.97 (0.80-1.18)	1.12 (0.81-1.55)
Family history of CAD [28 32]	EMS	2	1095	35 (16-50)	-	61 (48-80)	-	0.89 (0.73-1.50)	1.07 (0.69-1.39)	0.82 (0.50-1.92)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 4

eTable 4. Performance of risk factors in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

Risk factor	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Male gender [34]	GP	1	243	77 (61-86)	-	53 (46-60)	-	1.61 (1.34-1.93)	0.46 (0.38-0.55)	3.49 (1.67-7.23)
History of clinical vascular disease [34]	GP	1	243	49 (34-64)	-	58 (50-65)	-	1.15 (0.90-1.48)	0.89 (0.69-1.14)	1.30 (0.68-2.48)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 5.

eTable 5. Additional information about Point-of-Care (POC) analyzers							
Study	Setting	Manufacturer	Assay	Median duration (minutes)	Test result (minutes)	Succes rate (%)	Positive threshold
Van Dongen[2]	EMS	Roche Diagnostics	Troponin T	150	8 – 12	90	0.04 µg/L
Andersson [3]	GP	Roche Diagnostics	Troponin T	600	14	-	0.03 µg/L
Willemsen [4]	GP	FABPulous BV	H-FABP	-	-	-	4 µg/L
Nilsson [5]	GP	Roche Diagnostics	Troponin T	600	14	-	0.03 µg/L
Bruins Slot [6]	GP	Cardiodetect Rennesens GmbH	H-FABP	180	15	89	7 µg/L
Stopyra [7]	EMS	Abbott	Troponin I	-	8-10	-	0.08 µg/L
Svensson [8]	EMS	Cardiac Status Spectral Diagnostics Tests	Myoglobin CK-MB Troponin I	90	15	-	2.91 nmol/L 5 µg/L 0.1 µg/L
Stengaard [9]	EMS	Roche Diagnostics	Troponin T	70	12	90%	0.05 µg/L

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Rasmussen [10]	EMS	Roche Diagnostics	Troponin T	55	12	-	0.05 µg/L
Sagel[11]	EMS	Abbott	Troponin I	294	8-10	-	0.34 µg/L
Abbreviations: POC, Point-of-Care; HS, high sensitive; GP, General practitioner; EMS, Emergency Medical Services							

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eTable 6.

eTable 6. Study Characteristics					
Source	Study size	Incidence of ACS, No (%)	Clinical Setting	Diagnostic tests	Reference standard
Van Dongen et al.,[6] 2018	700	116 (17)	EMS	Male DM Obesity Family history of CAD Current smoker Previous AMI Previous PCI Previous CABG Previous TIA/Stroke Previous PAV POC- Troponin HEAR score HEART score	30-day MACE: ACS, death, PCI/CABG
Van Dongen et al.,[13] 2019	689	116 (17)	EMS	Core-lab PMHP score Hs-Troponin	30-day MACE: ACS, death, PCI/CABG
Ishak et al.,[14] 2016	1127	192 (17)	EMS	Modified HEART score	Final hospital diagnosis: NSTEMI or uAP
Van Dongen et al.,[15] 2018	700	96 (14)	EMS	HEART score	Final hospital diagnosis: NSTEMI or uAP
Bruins Slot et al.,[10] 2013	298	66 (22)	GP	H-FABP	Final hospital diagnosis: ACS
Andersson, et al.,[8] 2015	115	6 (5)	GP	Hs-Troponin POC-Troponin	Final hospital diagnosis: ACS
Schols et al.,[3] 2019	243	45 (19)	GP	Chest pain Duration of symptoms Worse with excercise Pressure pain Patients assumes cardiac origin of pain History of clinical vascular disease	Final hospital diagnosis: ACS

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				Male Pain not reproduced on palpation ECG GP immediately suspected a serious condition MHS GP probability assessment Combines MHS and GP probability assessment	
Willemsen et al.,[4] 2019	303	32 (11)	GP	H-FABP GP decision to refer CRS 1-4	Final hospital diagnosis: ACS
Nilsson et al.,[9] 2012	196	13 (7)	GP	POC-Troponin GP decision to refer GP decision to refer with POC-Troponin	Final hospital diagnosis: ACS
Anroedh et al.,[16] 2018	1421	144 (10)	EMS	ECG	Final hospital diagnosis: NSTEMI or uAP
Svensson et al.,[2] 2002	538	307 (57)	EMS	Pale Nausea Clammy AP Heart failure ECG Myoglobin CK-MB Troponin	Final hospital diagnosis: ACS
Stopyra et al.,[5] 2020	395	74 (19)	EMS	Male DM Obesity Family history of CAD Current smoker Previous AMI Previous PCI Previous CABG Previous TIA/Stroke	30-day MACE: ACS, death

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				Previous PAV Hypercholesterolemia Hypertension Prior coronary disease Heart failure PMHP score Core-lab PMHP score	
Stengaard et al.,[7] 2013	985	227 (23)	EMS	Male DM Current smoker Previous AMI Previous PCI Previous CABG Previous smoker POC-Troponin	Final hospital diagnosis: ACS
Sagel et al.,[12] 2021	435	53 (12)	EMS	preHEART score HEART score	3- and 7-day MACE: ACS or death
Rasmussen et al.,[11] 2017	17938	447 (2)	EMS	POC-Troponin	Final hospital diagnosis: NSTEMI
Abbreviations: ACS, Acute Coronary Syndrome; GP, General practitioner; EMS, Emergency Medical Services; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease; POC, Point-of-Care; HS, high sensitive; H-FABP; Heart-type Fatty Acid Binding Protein; HEART, History, ECG, Age, Risk Factors, Troponin; PMHP, prehospital modified HEART pathway; CRS, Combined Risk Scores; MHS, Marburg Heart Score; NSTEMI, non-ST-elevation myocard infarct; UAP, unstable angina pectoris; MACE, Major Adverse Cardiac Event					

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eTable 7.

eTable 7. Results of individual studies of symptoms						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Svensson et al.,[2] 2006	Chestpain	536	294	213	13	16
	Dyspnoe	536	150	124	157	105
	Nausea	536	114	66	193	163
Schols et al.,[3] 2019	Chestpain	243	39	176	6	22
	Worse with exertion	188	17	69	19	83
	Chestpain feels like pressure pain	212	31	154	8	19
	Patients assumes cardiac origin of pain	242	28	108	17	89
	Symptoms duration <1h	243	39	176	6	22
	Symptoms duration (1-24h)	243	23	15	22	83
	Symptoms duration (>24h)	243	19	63	26	135
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives						

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eTable 8.

eTable 8. Results of individual studies of risk factors						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Stopyra et al.,[5] 2020	Prior coronary disease	389	37	76	37	239
	Previous PAV	395	4	17	70	304
Van Dongen et al.,[6] 2018	Previous PAV	700	11	20	105	564
Svensson et al.,[2] 2003	Previous AMI	536	160	80	147	149
Stopyra et al.,[5] 2020	Previous AMI	589	28	43	46	472
Van Dongen et al.,[6] 2018	Previous AMI	700	34	116	82	468
Stengaard et al.,[7] 2013	Previous AMI	985	82	206	149	548
Stopyra et al.,[5] 2020	Previous CABG	390	8	27	66	289
Van Dongen et al.,[6] 2018	Previous CABG	700	14	52	102	532
Stengaard et al.,[7] 2013	Previous CABG	985	21	13	210	721
Stopyra et al.,[5] 2020	Previous PCI	395	19	78	55	243
Van Dongen et al.,[6] 2018	Previous PCI	700	42	133	74	451
Stengaard et al.,[7] 2013	Previous PCI	985	73	179	158	575
Stopyra et al.,[5] 2020	Previous TIA/Stroke	395	7	34	67	287
Van Dongen et al.,[6] 2018	Previous TIA/Stroke	700	11	32	105	552
Svensson et al.,[2] 2003	Current smoker	700	30	126	86	458
Stopyra et al.,[5] 2020	Current smoker	395	16	87	58	234
Van Dongen et al.,[6] 2018	Current smoker	700	30	126	86	458
Stengaard et al.,[7] 2013	Current smoker	1084	80	120	151	733
Svensson et al.,[2] 2003	Male gender	536	181	128	126	101
Stopyra et al.,[5] 2020	Male gender	395	43	142	31	179
Van Dongen et al.,[6] 2018	Male gender	700	88	313	28	271
Stengaard et al.,[7] 2013	Male gender	985	170	413	61	341
Svensson et al.,[2] 2003	AP	536	196	115	111	114

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Stopyra et al.,[5] 2020	Hypercholesterolemia	395	26	83	48	238
Van Dongen et al.,[6] 2018	Hypercholesterolemia	700	61	214	55	370
Stengaard et al.,[7] 2013	Hypercholesterolemia	985	194	624	37	130
Svensson et al.,[2] 2003	DM	536	58	32	249	197
Stopyra et al.,[5] 2020	DM	395	29	94	45	227
Van Dongen et al.,[6] 2018	DM	700	26	87	90	497
Stengaard et al.,[7] 2013	DM	985	38	143	193	611
Svensson et al.,[2] 2003	Heart failure	536	68	57	239	172
Stopyra et al.,[5] 2020	Heart failure	389	14	40	59	276
Svensson et al.,[2] 2003	Hypertension	536	107	57	200	172
Stopyra et al.,[5] 2020	Hypertension	389	55	201	18	115
Van Dongen et al.,[6] 2018	Hypertension	700	66	306	50	278
Stengaard et al.,[7] 2013	Hypertension	985	133	405	98	349
Stopyra et al.,[5] 2020	Obesity	383	29	152	45	157
Van Dongen et al.,[6] 2018	Obesity	700	27	113	89	471
Stengaard et al.,[7] 2013	Previous smoker	985	69	208	162	546
Stopyra et al.,[5] 2020	Family history of CAD	395	19	78	55	243
Van Dongen et al.,[6] 2018	Family history of CAD	700	47	277	69	307
Schols et al.,[3] 2019	Gender	243	34	93	11	105
	History of clinical vascular disease	243	22	84	23	114

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease; TP, true positives; FP, false positives; FN, false negatives, TN, true negatives

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eTable 9.

eTable 9. Results of individual studies of electrocardiogram findings						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Anroedh et al.,[16] 2018	Ischemic ECG	1421	69	457	75	820
Svensson et al.,[2] 2003	Ischemic ECG	536	52	126	255	103
Schols et al.,[3] 2019	Ischemic ECG	115	17	20	6	72
Svensson et al.,[2] 2003	Q-wave	536	33	1	274	228
	ST-depression	536	129	25	178	204
	T-wave inversion	536	55	16	252	213
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives						

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eTable 10.

eTable 10. Results of individual studies of biomarkers						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Svensson et al.,[2] 2003	POC-Troponin	536	21	2	286	227
Stengaard et al.,[7] 2013	POC-Troponin	924	73	34	146	671
Stopyra et al.,[5] 2020	POC-Troponin	395	17	11	57	310
Van Dongen et al.,[6] 2018	POC-Troponin	700	43	26	73	558
Rasmussen et al.,[11] 2019	POC-Troponin	17938	640	1175	1026	15097
Andersson et al.,[8] 2015	POC-Troponin	115	2	2	4	107
Nilsson et al.,[9] 2013	POC-Troponin	128	2	3	5	118
Willemsen et al.,[4] 2019	H-FABP	291	8	8	23	252
Bruins Slot et al.,[10] 2013	H-FABP	298	26	14	40	218
Svensson et al.,[2] 2003	CK-MB	536	37	7	270	222
	Myoglobin	536	25	9	282	220
Andersson et al.,[8] 2015	Hs-Troponin	115	5	26	1	83
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; POC, Point-of-Care; H-FABP; Heart-type Fatty Acid Binding Protein; HS, high-sensitive						

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eTable 11.

eTable 11. Results of individual studies of Combined Risk Scores						
Study	Characteristic	Risk	TP	FP	FN	TN
Stopyra et al.,[5] 2020	Core-lab PMHP	High	50	25		
		Intermediate	24	172		
		Low	0	172		
	PMHP	High	17	11		
		Intermediate	50	193		
		Low	7	117		
Sagel et al.,[12] 2021	preHEART	High	55	56		
		Intermediate	117	797		
		Low	4	614		
	HEART	High	67	122		
		Intermediate	51	648		
		Low	5	315		
Ishak et al.,[14] 2018	Modified HEART	High	116	114		
		Intermediate	76	418		
		Low	0	403		
Willemsen et al.,[4] 2019	CRS 1	High - Low	28	130	4	141
	CRS 2	High - Low	26	130	6	141
	CRS 3	High - low	25	121	7	150
	CRS 4	High - Low	22	120	10	151
Schols et al.,[3] 2019	MHS	Cut-off value 2	27	84	9	66
	MHS	Cut- off value 1	34	126	2	24
	GP probability assessment	High - Low	36	116	6	82
	Combined MHS & GP probability assessment	High - Low	36	115	0	35
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; CRS, Combined Risk Scores; HEAR, History, ECG, Age, Risk Factors; POC, Point-of-Care; H-FABP, Heart-type Fatty Acid Binding Protein; MHS, Marburg Heart Score; GP, general practitioner						

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eTable 12.

eTable 12. Results of individual studies of general practitioner decision making						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Willemsen et al.,[4] 2019	GP decision to refer	303	24	88	8	183
Nilsson et al.,[9] 2013	GP decision to refer	68	6	23	0	39
	GP decision to refer with POC-Troponin	128	5	27	2	94
Schols et al.,[3] 2019	GP immediately suspected a serious condition	243	26	94	19	104
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; GP, General Practitioner; POC, Point-of-Care						

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References

1. Whiting PF, Weswood ME, Rutjes AW, et al. Evaluation of QUADAS, a tool for the quality assessment of diagnostic accuracy studies. *BMC Med Res Methodol* 2006;6:9. doi: 10.1186/1471-2288-6-9 [published Online First: 2006/03/08]
2. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Value of Prehospital Troponin Assessment in Suspected Non-ST-Elevation Acute Coronary Syndrome. *Am J Cardiol* 2018;122(10):1610-16. doi: 10.1016/j.amjcard.2018.07.037 [published Online First: 2018/09/29]
3. Andersson PO, Karlsson JE, Landberg E, et al. Consequences of high-sensitivity troponin T testing applied in a primary care population with chest pain compared with a commercially available point-of-care troponin T analysis: an observational prospective study. *BMC Res Notes* 2015;8:210. doi: 10.1186/s13104-015-1174-0 [published Online First: 2015/06/04]
4. Willemsen RT, Winkens B, Kietselaer BL, et al. Evaluating possible acute coronary syndrome in primary care: the value of signs, symptoms, and plasma heart-type fatty acid-binding protein (H-FABP). A diagnostic study. *BJGP Open* 2019;3(3) doi: 10.3399/bjgpopen19X101652 [published Online First: 2019/10/04]
5. Nilsson S, Andersson PO, Borgquist L, et al. Point-of-Care Troponin T Testing in the Management of Patients with Chest Pain in the Swedish Primary Care. *Int J Family Med* 2013;2013:532093. doi: 10.1155/2013/532093 [published Online First: 2013/02/01]
6. Bruins Slot MH, Rutten FH, van der Heijden GJ, et al. Diagnostic value of a heart-type fatty acid-binding protein (H-FABP) bedside test in suspected acute coronary syndrome in primary care. *Int J Cardiol* 2013;168(2):1485-9. doi: 10.1016/j.ijcard.2012.12.050 [published Online First: 2013/01/22]
7. Stopyra JP, Snavely AC, Smith LM, et al. Prehospital use of a modified HEART Pathway and point-of-care troponin to predict cardiovascular events. *PLoS One* 2020;15(10):e0239460. doi: 10.1371/journal.pone.0239460 [published Online First: 2020/10/08]
8. Svensson L, Isaksson L, Axelsson C, et al. Predictors of myocardial damage prior to hospital admission among patients with acute chest pain or other symptoms raising a suspicion of acute coronary syndrome. *Coron Artery Dis* 2003;14(3):225-31. doi: 10.1097/01.mca.0000063503.13456.0d [published Online First: 2003/04/19]
9. Stengaard C, Sorensen JT, Ladefoged SA, et al. Quantitative point-of-care troponin T measurement for diagnosis and prognosis in patients with a suspected acute myocardial infarction. *Am J Cardiol* 2013;112(9):1361-6. doi: 10.1016/j.amjcard.2013.06.026 [published Online First: 2013/08/21]
10. Rasmussen MB, Stengaard C, Sorensen JT, et al. Predictive value of routine point-of-care cardiac troponin T measurement for prehospital diagnosis and risk-stratification in patients with suspected acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;8(4):299-308. doi: 10.1177/2048872617745893 [published Online First: 2017/12/05]
11. Sagel DC, Vlaar PJ, Waardenburg I, et al. Pre-hospital risk stratification in patients with undifferentiated chest pain. *Emerg Med J. 2021.*

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Research Reporting Guideline checklist

Demandt JPA, Zelis JM, Koks A, et al. Prehospital Risk Assessment In Suspected Acute Coronary Syndrome Patients – A Systematic Review And Meta-analysis

PRISMA-DTA Checklist

Section/topic	#	PRISMA-DTA Checklist Item	Reported on page #
TITLE / ABSTRACT			
Title	1	Identify the report as a systematic review (+/- meta-analysis) of diagnostic test accuracy (DTA) studies.	1
Abstract	2	Abstract: See PRISMA-DTA for abstracts.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5/6
Clinical role of index test	D1	State the scientific and clinical background, including the intended use and clinical role of the index test, and if applicable, the rationale for minimally acceptable test accuracy (or minimum difference in accuracy for comparative design).	5/6
Objectives	4	Provide an explicit statement of question(s) being addressed in terms of participants, index test(s), and target condition(s).	5/6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	7
Eligibility criteria	6	Specify study characteristics (participants, setting, index test(s), reference standard(s), target condition(s), and study design) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7/8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full search strategies for all electronic databases and other sources searched, including any limits used, such that they could be repeated.	Supplementary
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Definitions for data extraction	11	Provide definitions used in data extraction and classifications of target condition(s), index test(s),	8

		reference standard(s) and other characteristics (e.g. study design, clinical setting).	
Risk of bias and applicability	12	Describe methods used for assessing risk of bias in individual studies and concerns regarding the applicability to the review question.	7
Diagnostic accuracy measures	13	State the principal diagnostic accuracy measure(s) reported (e.g. sensitivity, specificity) and state the unit of assessment (e.g. per-patient, per-lesion).	8
Synthesis of results	14	Describe methods of handling data, combining results of studies and describing variability between studies. This could include, but is not limited to: a) handling of multiple definitions of target condition. b) handling of multiple thresholds of test positivity, c) handling multiple index test readers, d) handling of indeterminate test results, e) grouping and comparing tests, f) handling of different reference standards	8

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Section/topic	#	PRISMA-DTA Checklist Item	Reported on page #
Meta-analysis	D2	Report the statistical methods used for meta-analyses, if performed.	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	-
RESULTS			
Study selection	17	Provide numbers of studies screened, assessed for eligibility, included in the review (and included in meta-analysis, if applicable) with reasons for exclusions at each stage, ideally with a flow diagram.	10/Supplementary
Study characteristics	18	For each included study provide citations and present key characteristics including: a) participant characteristics (presentation, prior testing), b) clinical setting, c) study design, d) target condition definition, e) index test, f) reference standard, g) sample size, h) funding sources	Supplementary
Risk of bias and applicability	19	Present evaluation of risk of bias and concerns regarding applicability for each study.	Supplementary
Results of individual studies	20	For each analysis in each study (e.g. unique combination of index test, reference standard, and positivity threshold) report 2x2 data (TP, FP, FN, TN) with estimates of diagnostic accuracy and confidence intervals, ideally with a forest or receiver operator characteristic (ROC) plot.	Supplementary
Synthesis of results	21	Describe test accuracy, including variability; if meta-analysis was done, include results and confidence intervals.	10-21
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression; analysis of index test: failure rates, proportion of inconclusive results, adverse events).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence.	23-26
Limitations	25	Discuss limitations from included studies (e.g. risk of bias and concerns regarding applicability) and from the review process (e.g. incomplete retrieval of identified research).	26/27
Conclusions	26	Provide a general interpretation of the results in the context of other evidence. Discuss implications for future research and clinical practice (e.g. the intended use and clinical role of the index test).	23-27
FUNDING			
Funding	27	For the systematic review, describe the sources of funding and other support and the role of the funders.	29

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Prehospital risk assessment in patients suspected of Non-ST-segment elevation Acute Coronary Syndrome – A systematic review and meta-analysis

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TITLE PAGE

Prehospital risk assessment in patients suspected of Non-ST-segment elevation Acute Coronary Syndrome – A systematic review and meta-analysis

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Demandt et al. Prehospital risk assessment in ACS_Revised

ABSTRACT (399 words)

OBJECTIVE

To review, inventory and compare available diagnostic tools and investigate which tool has the best performance for prehospital risk assessment in patients suspected of non-ST-segment elevation Acute Coronary Syndrome (NSTE-ACS).

METHODS

Systematic review and meta-analysis. MEDLINE and EMBASE were searched up till April 1, 2021. Prospective studies with patients, suspected of NSTE-ACS, presenting in the primary care setting or by Emergency Medical Services (EMS) were included. The most important exclusion criteria were studies including only patients with ST-elevation myocardial infarction and studies before 1995, the pre-troponin era. The primary end point was the final hospital discharge diagnosis of NSTE-ACS or Major Adverse Cardiac Events (MACE) within 6 weeks. Risk of bias was evaluated by the Quality Assessment of Diagnostic Accuracy Studies (QUADAS)-Criteria.

MAIN OUTCOME AND MEASURES

Sensitivity, specificity, and, likelihood ratio of findings for risk stratification in patients suspected of NSTE-ACS.

RESULTS

In total 15 prospective studies were included, these studies reflected in total 26083 patients. No specific variables related to symptoms, physical examination or risk factors were useful in risk stratification for NSTE-ACS diagnosis. The most useful electrocardiographic finding was ST-segment depression (LR+ 3.85 [95% CI, 2.58 to 5.76]). Point-of-care Troponin was found to be a strong predictor for NSTE-ACS in primary care (LR+ 14.16 [95% CI, 4.28 to 46.90] and EMS setting (LR+ 6.16 [95% CI, 5.02 to 7.57]). Combined risk scores were the best for risk assessment in an NSTE-ACS. From the combined risk scores that can be used immediately in a prehospital setting, the PreHEART score,

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a validated combined risk score for prehospital use, derived from the HEART score (History, Electrocardiogram, Age, Risk factors, Troponin), was most useful for risk stratification in patients with NSTE-ACS (LR+ 8.19 [95% CI, 5.47 to 12.26]) and for identifying patients without ACS (LR- 0.05 [95% CI, 0.02 to 0.15]).

DISCUSSION

Important study limitations were verification bias and heterogeneity between studies. In the prehospital setting several diagnostic tools have been reported which could improve risk stratification, triage and, early treatment in patients suspected for NSTE-ACS. On-site assessment of Troponin and combined risk scores derived from the HEART score are strong predictors. These results support further studies to investigate the impact of these new tools on logistics and clinical outcome.

FUNDING

This study is funded by ZonMw, the Dutch Organisation for Health Research and Development

REGISTRATION

This meta-analysis was published for registration in PROSPERO prior to starting (CRD York, CRD42021254122).

KEYWORDS

NSTE-ACS, undifferentiated chest pain, prehospital, early triage, risk stratification, diagnostic tools, combined risk scores

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STRENGTHS AND LIMITATIONS

1. This systematic review and meta-analysis includes all currently available prospective data on the subject of pre-hospital risk assessment in NSTE-ACS patients.
2. Available patient characteristics and diagnostic tools (including recently developed combined risks scores and point-of-care biomarkers) are reviewed and compared in both the setting of primary care as well as emergency medical care.
3. Despite strict inclusion and exclusion criteria, heterogeneity was present in our review.
4. For none of the included prehospital diagnostic tools there were prospective intervention and/or randomized controlled studies available.

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INTRODUCTION

The suspicion of an Acute Coronary Syndrome (ACS) is one of the main reasons for consultation of emergency medical care.[1-4] Depending on national and regional protocols, patients who are suspected of having an ACS are first seen in primary care by a general practitioner (GP) or by the Emergency Medical Services (EMS). The decision to admit the suspected ACS patient directly to a tertiary intervention center, or to transport the patient to a local hospital for observation results from varying risk stratification methods. In general, this prehospital risk assessment and triage is mainly based on the patient's medical history and physical examination. None or only a limited number of diagnostic tools are used in this process. Often, only an electrocardiogram (ECG) is made by the EMS to rule out a ST-segment elevation myocardial infarction (STEMI). Consequently, no adequate prehospital risk assessment and triage is performed in those without ST-segment elevation on their ECG (NSTE-ACS) and are transferred to a local hospital without onsite interventional options for further diagnostic evaluation.[2]

In the recent years, in-hospital diagnostic algorithm for suspected NSTE-ACS have been developed and improved. The implementation of combined risk scores such as the HEART and TIMI scores and new sensitive biomarker assays have substantially improved risk stratification, reduced the delay to diagnosis, shortened stays in the emergency departments and lowered health care costs.[5,6] Further improvement of care and cost reduction could be possible when these diagnostic algorithms would be applicable in the prehospital phase. The recent development of point-of-care (POC) analyzers for on-site biomarker assessment have accelerated these possibilities of prehospital risk assessment in suspected NSTE-ACS patients. Improving early risk stratification by GP or ambulance paramedics is important for several medical and economic reasons. Most importantly, due to the absence of reliable prehospital diagnostic tools, an NSTE-ACS is not always recognized, leading to a missed diagnosis in around 2-5% of consultations.[7] Additionally, the latest European guidelines recommend

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that high-risk NSTE-ACS patients receive an early invasive strategy within 24 hours.[8] It is well known that this is associated with shorter ischemic times and subsequent improved clinical outcomes.[9-11]

On the other hand, the use of additional diagnostic tools could help to definitively rule out an NSTE-ACS in the prehospital setting in chest pain patients with a low suspicion for NSTE-ACS. This is important as only 11-21% of these patients presenting to an emergency department are finally diagnosed with an NSTE-ACS, in the remaining majority generally no life-threatening condition is found.[12-14] Early risk assessment in these patients with low suspicion could significantly reduce the need of unnecessary diagnostics, referrals to hospitals, and therefore reduce health care cost.

Recently, several studies have been performed to investigate several known and new diagnostic tools for prehospital risk assessment in suspected ACS patients.[15-17] We performed a systematic review and meta-analysis of available prospective data to summarize and compare current available and new diagnostic tools for prehospital risk assessment in patients suspected of NSTE-ACS.

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METHODS

Study Selection and Data Extraction

This meta-analysis was published for registration in PROSPERO prior to starting (CRD York, CRD42021254122).

We performed searches in MEDLINE and EMBASE for all published articles up till April 1, 2021 . After identifying articles, we reviewed references from appropriate articles to identify additional references for this systematic review. Full search is shown in eAppendix 1.

Titles and abstract for all articles were screened by the primary author (J.D.) and decisions were checked by a second reviewer (P.J.V.). If the author identified the article as potentially suitable for inclusion, the full text of the article was reviewed in detail by two authors (J.D. and P.J.V.) and selected whether the inclusion criteria were met. Disagreements between reviewers were resolved by the decision of a third independent reviewer. If data were sufficient to generate a 2 x 2 table, the data was independently extracted, and the methodological quality and eligibility were determined. Bias was evaluated by the Quality Assessment of Diagnostic Accuracy Studies (QUADAS)-Criteria.[18]

We included studies that met the following inclusion criteria: (1) patients suspected for NSTE-ACS; (2) prospective study; (3) original data ; (4) presenting prehospital (primary care, EMS); (5) prehospital risk assessment or triage; (6) outcome data on in-hospital ACS or MACE within 6 weeks after presentation; (7) article published in English, French, or German.

The objective of this review was to investigate which patient characteristics and diagnostic tools were used for early risk assessment in an undifferentiated prehospital population suspected of NSTE-ACS, we excluded studies enrolling a specific subpopulation from the general ACS population, studies with less than 100 patients, and studies enrolling only STEMI patients. Further, we excluded studies from which the data was incomplete, or the endpoints were not focused on diagnostic performance. Studies before 1995, the pre-troponin era, were excluded as well.

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End Points and Definitions

The primary end point was the final hospital discharge diagnosis of NSTE-ACS by the treating physician or Major Adverse Cardiac Event (MACE) within 6 weeks. NSTE-ACS refers to non-ST-elevation myocardial infarction (NSTEMI) or unstable angina pectoris (UAP). MACE was defined in the studies and included at least myocardial infarction.

Data and statistical analysis

For included studies individual data of sample size, number of true positive, true negative, false positive and false negative were extracted separately by two authors (J.D. and P.J.V.). When possible, we excluded the STEMI patients from the data extraction. Sensitivity, specificity, likelihood ratios (LRs), and diagnostics odds ratio (dOR) were calculated from the extracted data by two authors (J.D. and J.Z.). The statistical analysis was checked by a third author (M.V.). Extracted data included symptoms, physical examination, risk factors, ECG, biomarkers and Combined Risk Scores (CRS).

Since LRs is an appropriate way to measure and express diagnostic accuracy, our main focus was on this diagnostic measurement.[19,20] An LR around 1.0 has no proven diagnostic value. A positive likelihood ratio (LR+) between 1 and 2 as a minimal predictor for making ACS more likely, a LR+ of >2 as a weak predictor for ACS, a LR+ >5 as a moderate to strong predictor and a LR >10 as a very strong predictor. A negative likelihood ratio (LR-) between 0.5 and 1 as a minimal predictor to make ACS less likely, a LR- of <0.5 as a weak predictor, LR- < 0.2 as a moderate to strong predictor and a LR- <0.1 as a very strong predictor.[21]

Summary data for dichotomous findings are reported as weighted average when a finding was evaluated in 2 studies. When findings were reported in 3 or more studies the analyses were performed using R version 4.0.4 (R Foundation for Statistical Computing, Vienna, Austria). Sensitivity, specificity, and the diagnostic odds ratio analyses were performed using the 'meta' package as was heterogeneity. Likelihood ratios and bivariate analyses were performed using the 'mada' package.[22]

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Statistical heterogeneity was reported as the I^2 for the sensitivity and specificity when findings were evaluated in 3 studies or more. We considered values between 0% and 30% as low heterogeneity, between 30% and 60% as moderate heterogeneity and 60% or more as considerable heterogeneity.[23]

Patient and public involvement

Patient and public were not involved.

Statement of Ethics approval

It was not necessary to engage an ethics committee for this study-level systematic review and meta-analysis.

Data sharing statement

Not applicable

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RESULTS

Of the 1817 unique articles a total of 15 prospective studies met our inclusion criteria and were included in this systematic review **(eFigure 1)**.[15-17 24-35] These studies reflected a total of 26083 patients. Five studies were performed in setting of primary care by the GP and 10 in the setting of the EMS. Rates of final NSTE-ACS diagnosis ranged in primary care from 5% to 22% (median 11% [interquartile range (IQR), 6%-20%]. In the EMS setting the rates of final ACS diagnosis ranged from 8% to 57% (median 17% [IQR, 12%-20%].

Risk of bias for each included study was scored with the QUADAS tool (**eAppendix 2)**. Data on 92 variables related to symptoms, physical examination, risk factors, ECG, biomarkers, and risk scores were extracted from the articles. The total number of studies providing data, the total number of patients, sensitivity, specificity, LR+, LR- and dOR are listed in the supplementary material. All data relevant to the systematic review and meta-analysis are uploaded as supplementary material. **(eAppendix 3 – 10).**

Symptoms

Symptoms were not predictive for ACS in primary care or on the EMS setting **(eTable 1)**. The absence of having chest pain in the ambulance was a minimal predictor for making ACS less likely (LR- 0.61 [0.58-0.63]). Duration of symptoms in primary care until presentation were not appropriate for risk assessment in ACS.

Physical examination

Pain not reproducible by palpation (LR+ 1.10 [1.02-1.20]) and pallor (LR+ 1.17 [1.00-1.38]) were minimally predictive for making ACS more likely **(eTable 2)**. Reproducing the pain by palpation is a weak predictor for making ACS less likely (LR- 0.47 [0.43-0.51]) in EMS setting.

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Cardiovascular risk factors.

Of all the cardiovascular risk factors in EMS setting, history of prior coronary disease (LR+ 2.07 [1.58-2.71]) was the best predictor to make ACS more likely **(eTable 3)**. Male gender (LR+ 1.28 [1.12-1.45], history of angina pectoris (AP) (LR+ 1.27 [1.08-1.49]) or previous AMI (LR+ 1.83 [1.21-2.80]), CABG (LR+ 1.60 [1.14-2.24] or PCI (LR+ 1.49 [1.24-1.78]) were minimal predictors for ACS. For identifying patients less likely to have an ACS, no risk factor when absent conferred an LR- of 0.5 or lower. In primary care male gender (LR+ 1.61 [1.34-1.93]) was the only risk factor for making ACS more likely **(eTable 4)**.

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ECG

A total of 3 prospective studies investigated the value of ECG, 1 in primary care and 2 in EMS. [24,25,34] In all 3 studies the ECG was categorized in ischemic ECG versus abnormal and/or non-ischemic ECG. In only one study the exact definition of an ischemic ECG was mentioned: ST-deviation ≥800 µV. [24] Interpreting an ECG as an ischemic ECG is predictive for ACS in both the EMS setting (LR+ 2.23 [1.11-5.30]) as in primary care (LR+ 3.4 [2.15-5.37]). Classifying the ECG as non-ischemic is a weak predictor for making ACS less likely in primary care (LR- 0.33 [0.17-0.67]) and a minimal predictor in the EMS setting (LR- 0.57 [0.33-0.95]) (**Table 1)**. In terms of individual characteristics of the ECG, a Q-wave is a very strong predictor for ACS (LR+ 24.62 [3.38-179.20]. ST-depression (LR+ 3.85 [2.58-5.76]) is a better predictor for ACS than T-wave inversion (LR + 2.56 [1.50-4.39]) **(Table 1).**

Table 1. Performance of the ECG for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Ischemic ECG[b] [24,25]	EMS	2	1959	58 (45-71)	-	74 (59-85)	-	2.23 (1.11-5.30)	0.57 (0.33-0.95)	4.03 (0.70-23.42)
Ischemic ECG[b] [34]	GP	1	243	74 (53-88)	-	78 (69-84)	-	3.4 (2.15-5.37)	0.33 (0.17-0.67)	10.20 (3.55-29.28)
Q-wave [25]	EMS	1	541	11 (8-15)	-	100 (98-100)	-	25.36 (3.49-184.47)	0.89 (0.12-6.50)	28.39 (3.90-209.01)

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ST-depression [25]	EMS	1	541	42 (36-48)	-	89 (84-93)	-	3.85 (2.58-5.76)	0.65 (0.43-0.97)	5.91 (3.68-9.49)
T-wave inversion [25]	EMS	1	541	18 (14-23)	-	93 (89-96)	-	2.56 (1.50-4.39)	0.88 (0.52-1.51)	2.91 (1.62-5.22)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

[b] Definition of ischemic ECG given in 1 study only; ST-deviation ≥800 µV

Biomarkers

A total of 9 prospective studies investigated the value of biomarkers, 4 in primary care and 5 in EMS.[17 24-29 33 34] All blood samples for biomarkers were drawn in the prehospital setting. Most samples were analyzed directly prehospital by POC analyzers, but in 1 study the blood sample was analyzed after arrival in the hospital.[27] In EMS setting, median time of onset of complaint to biomarker assessment was 80 minutes (IQR 59–135 minutes) and in primary care the median was 600 (IQR 180-600 minutes). Further details on the prehospital analyzed biomarkers are shown in **eTable 5.**

Of the analyzed biomarkers in primary care the Heart-type Fatty Acid Binding Protein (H-FABP) (LR+ 7.84 [3.39-20.76]) was moderate to strong for predicting ACS **(Table 2).** POC-Troponin (LR + 14.16 [4.28-46.90]) was a very strong predictor for ACS. Both were not useful to make an ACS less likely (LR- 0.68 [0.31-1.91] vs LR- 0.71 [0.49-1.02]) **(Table 3)**.

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In the EMS setting, POC-Troponin (LR+ 6.16 [5.02-7.57]) was a moderate to strong predictor for making ACS more likely that could be interpreted immediately. There was no difference between blood sampled by POC in the EMS and blood analyzed in-hospital.

Table 2. Performance of the biomarkers for risk assessment in non-ST-elevation Acute Coronary Syndrome

Assay	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
H-FABP [26 28]	GP	2	601	35 (12-52)	-	96 (90-99)	-	7.84 (3.39-20.76)	0.68 (0.31-1.90)	11.53 (3.76-31.91)
Hs-Troponin T [27]	GP	1	115	83 (36-100)	-	76 (67-84)	-	3.49 (2.40-5.09)	0.22 (0.15-0.32)	15.97 (1.79-142.88)
Myoglobin [25]	EMS	1	536	8 (5-12)	-	96 (93-98)	-	2.07 (0.98-4.38)	0.96 (0.45-2.02)	2.17 (0.99-4.74)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; H-FABP; Heart-type Fatty Acid Binding Protein; HS, high-sensitive

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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Table 3. Performance of the POC-Troponin for risk assessment in non-ST-elevation Acute Coronary Syndrome

	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
POC-Troponin [15 17 25,30, 33]	EMS	5	21484	25 (14-40)	96%	96 (94-98)	85%	6.16 (5.02-7.57)	0.75 (0.62-0.92)	8.27 (7.42-9.21)
POC-Troponin [27,29]	GP	2	311	31 (12-59)	-	98 (95-99)	-	14.16 (4.28-46.90)	0.71 (0.49-1.02)	20.01 (4.56-87.88)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; POC, Point-of-Care

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

Combined Risk Scores

Combined risk scores were investigated in 8 studies, 2 in primary care[28 34] and 6 in EMS[15 16 31-33 35]. In the EMS setting the used combined risk scores were all derived from the well-known HEART score.[5] In total 5 different combined risk scores were evaluated in this systematic review **(Table 4)**. There were minimal differences between the risk scores. These were due to different positivity threshold, risk factors and Troponin analyzers. In 2 studies data and blood were drawn prehospital but analyzed after arrival in-hospital.

In the EMS setting, all the combined risk scores were strong predictors for ACS when classified in the high-risk group. Between the risk scores there was no statistically significant difference **(Table 4).**

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In the low-risk group, the modified HEART performed the best in excluding ACS (LR-0.00). Of the combined risk scores that can be interpreted directly prehospital, preHEART is a very strong predictor in making ACS less likely (LR- (0.05 [0.02-0.15].

In all primary care studies, the combined risk scores classified patients in 2 risk groups (low vs intermediate/high). [28 34] In the EMS setting this was only the case with the HEAR score and was a weak predictor for making ACS less likely (LR- 0.40 [0.38-0.43]).[33] The Marburg Heart Score (MHS) showed a significant better performance in making ACS less likely when the threshold value was lowered from 2 to 1 (LR- 0.57 [0.47-0.68] vs LR- 0.35 [0.32-0.38]) **(Table 5)**. Alternative decision tools used in primary care, were the CRS. Scoring a positive answer to the elements, resulted in a point that was calculated into a final score. This final score classified the patient into a low risk group or intermediate-high risk group. CRS 1 had the best performance in making ACS less likely (LR- 0.24 [0.21-0.28]) and outperformed the MHS.

Compared to GP decision-making without use of a combined risk score (LR- 0.32 [0.00-0.46]) there is no improvement that makes ACS less likely **(Table 6)**.

Making an ACS more likely, decision making by the GP to refer to the hospital, without using a POC-troponin (LR+ 2.37 [1.95-3.73]) and using POC-Troponin (LR+ 3.20 [2.14-4.80]) outperformed the combined risk scores.

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Table 4. Performance of the Combined Risk Scores in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome				
Risk Level	**Studies**	**Threshold**	**LR (95-CI)**	**Troponin analyzer**
High				
Core-lab PMHP [15]	1	Positive Troponin	8.68 (5.04-14.93)	In-hospital laboratorium
preHEART [35]	1	7-10	8.19 (5.47-12.26)	POC
PMHP [15]	1	Positive Troponin	6.70 (3.01-14.91)	POC
Modified HEART [31]	1	7-10	4.96 (3.62-6.79)	In-hospital laboratorium
HEART [35]	1	7-10	4.84 (3.41-6.89)	POC
Intermediate				
Core-lab PMHP [15]	1	≥4	0.61 (0.35-1.04)	In-hospital laboratorium
preHEART [35]	1	4-6	1.22 (0.95-1.57)	POC

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PMHP [15]	1	≥4	1.12 (0.74-1.70)	POC
Modified HEART [31]	1	4-6	0.89 (0.66-1.18)	Lab
HEART [35]	1	4-6	0.69 (0.49-0.98)	POC
Low				
Core-lab PMHP [15 16]	2	0-3	0.08 (0-0.17)	In-hospital laboratorium
preHEART [35]	1	0-3	0.05 (0.02-0.15)	POC
PMHP [15]	1	0-3	0.26 (0.11-0.59)	POC
Modified HEART [31]	1	0-3	0.00	In-hospital laboratorium
HEART [32 35]	2	0-3	0.14 (0.13-0.16)	POC
Abbreviations: 95-CI, 95% confidence interval; LR, likelihood ratio; POC. Point-of-Care; HEART, History, ECG, Age, Risk Factors, Troponin; PMHP, prehospital modified HEART pathway				

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Table 5. Performance of the Combined Risk Scores in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

CRS	Studies	Threshold for intermediate-high risk group	LR+ (95-CI)	Threshold for low-risk group	LR- (95-CI)	Setting	Elements of CRS
CRS 1 [28]	1	2-6	1.82 (1.59-2.10)	0-1	0.24 (0.21-0.28)	GP	• ST-depression • ST-elevation • Dyspnoe • Feeling of pressure chest pain • Absent laterale chest pain left • POC H-FABP +
CRS 2 [28]	1	2-5	1.69 (1.47-1.96)	0-1	0.36 (0.31-0.42)	GP	• ST-depression • ST-elevation • Dyspnoe • Feeling of pressure chest pain • Absent lateral chest pain left
CRS 3 [28]	1	2-4	1.75 (1.49-2.06)	0-1	0.40 (0.34-0.46)	GP	• Dyspnoe

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							• Feeling of pressure chest pain • Absent lateral chest pain left • POC H-FABP +
CRS 4 [28]	1	2-3	1.55 (1.30-1.85)	0-1	0.56 (0.47-0.67)	GP	• Dyspnoe • Feeling of pressure chest pain • Absent lateral chest pain left
MHS [34]	1	≥3	1.34 (1.11-1.61)	0-2	0.57 (0.47-0.68)	GP	
MHS [34]	1	≥2	1.12 (1.04-1.22)	0-1	0.35 (0.32-0.38)	GP	
GP probability assessment [34]	1	≥6	1.48 (1.29-1.70)	0-5	0.32 (0.28-0.37)	GP	
Combined MHS & GP probability assessment [34]	1	1-2	1.30 (1.19-1.42)	0	0	GP	
Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; CRS, Combined Risk Scores; HEAR, History, ECG, Age, Risk Factors; POC, Point-of-Care; H-FABP, Heart-type Fatty Acid Binding Protein; MHS, Marburg Heart Score; GP, general practitioner							

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Table 6. Performance of GP decision making for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
GP decision to refer [28 29]	2	371	79 (54-100)	-	67 (50-75)	-	2.37 (1.95-3.73)	0.32 (0.00-0.46)	7.50 (0-14.45)
GP decision to refer with POC-Troponin [29]	1	128	71 (29-96)	-	78 (69-85)	-	3.20 (2.14-4.80)	0.37 (0.25-0.55)	8.70 (1.60-47.40)
GP immediately suspected a serious condition [34]	1	243	58 (42-72)	-	53 (45-60)	-	1.22 (0.98-1.51)	0.80 (0.65-1.00)	1.51 (0.79-2.91)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; GP, General Practitioner; POC, Point-of-Care

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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Alternative diagnosis

In our systematic review 3 studies in the EMS setting described which alternative diagnoses were possible. From these, 2 studies noted the predictive value of POC-Troponin for other diseases[30,17], 1 study noted which other causes for chest pain could be classified as low-risk.[31] POC-Troponin is a moderate to strong predictor for other cardiac related diseases like myocarditis (LR+ 8.89 [8.29 – 9.54]), decompensated heart failure (LR+ 5.26 [1.88-6.03]), valve disorders/endocarditis (LR+ 3.29 [2.93-3.69]) and cardiomyopathy (LR+ 4.80 [4.36-5.29]). For non-cardiac diseases POC-Troponin is a weak to moderate predictor to make pulmonary embolism (LR+ 3.81 [1.38-4.46]), pneumonia (LR+ 2.41 [2.13-2.74]) and diseases of the genitourinary system more likely (LR+ 4.07 [2.66-4.51]). A normal POC-Troponin value makes a myocarditis less likely (LR- 0.40 [0.37-0.42]) but is not a strong predictor for other diagnoses. In the modified HEART, 403 patients were classified as low risk for ACS. In this group 2 serious non-cardiac diseased (pulmonary embolism, acute pancreatitis) were found.

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DISCUSSION

In this systematic review, we evaluated 15 prospective studies including 26.083 patients to identify patient characteristics and diagnostic tools for prehospital risk assessment in suspected NSTE-ACS patients. The main finding is that in the prehospital setting several new diagnostic tools are available which could improve early triage and risk assessment in suspected NSTE-ACS patients, in particular on-site assessment of biomarkers and combined risk scores derived from the HEART score were very strong for risk assessment in patients with NSTE-ACS.

Symptoms, physical examination, and risk factors

In accordance with previous analyses, in the present study no specific variables related to symptoms and physical examination were relevant predictors for an ACS diagnosis. The predictive value of some cardiovascular risk factors in the prehospital setting were as a single variable weak at best. A possible explanation is that in the prehospital setting less objective information is present on previous history of a patient. This was indeed corroborated in a recent analysis comparing prehospital and in-hospital calculated HEART scores.[16] In this study disagreement between hospital and prehospital HEART score risk classifications was found in approximately 25% of patients, mainly by different scoring of history and cardiovascular risk factors.

ECG

In patients with suspected ACS, the resting 12-lead ECG is the first-line diagnostic tool. Our results indicate that in the prehospital setting only a minority of the studies provided detailed description of the ECG criteria and used definitions. Moreover, the ECG was only a moderate predictor for NSTE-ACS. We expect that a more detailed analysis of ECGs, training of EMS personnel or GPs could improve the performance of ECG in the prehospital setting.

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Biomarkers

The advent of POC analyzers facilitates the use of established biomarkers in the prehospital setting. Current guidelines state that measurement of the biomarker Troponin is mandatory in all suspected ACS patients.[36] Although the included studies used normal Troponin I and T assays, the first high-sensitivity assays also became available recently for POC analyzers**.** This can increase diagnostic accuracy and reduce the time to diagnosis.[37] Depending on the type of assay used, assessment of this biomarker is already reliable 3-6 hours after onset of symptoms. The duration of symptoms could differ between different settings and populations. We found that duration of symptoms often is much longer in the primary care setting compared to the EMS setting. This explains why the negative predictive value of POC troponin is much higher in the primary care setting than in EMS setting. Consequently, symptom duration could indeed be too short to rule out an ACS based on a single troponin assessment in the EMS setting. However, the use of POC troponin in a combined risk score could offer a solution to lower the risk of false negatives. This is supported by the results from our analyses which clearly show that the combined risk scores are superior to a single troponin for risk assessment in a patient suspected for NSTE-ACS.

Two primary care studies[26 28] examined the diagnostic value of H-FABP, a small protein that is common in cardiac muscle and is released within 2 hours after the onset of ischemia.[38] Although there are no prehospital studies that compare Troponin with H-FABP, several in-hospital studies have compared these biomarkers. Despite of the early release, these studies showed that the predictive value of the H-FABP is moderate when assessed within the time frame of 6 hours after onset of symptoms. Compared to POC-Troponin, stand-alone H-FABP showed inferior diagnostic performance.[39,40,41]

Combined risk scores

International guidelines advise the use of combined risk scores at emergency departments.[36] Several scores are available that, generally combine elements of history, cardiovascular risk factors, ECG, and

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the biomarker troponin. The advent of POC analyzers makes it possible to use these risk scores in the prehospital setting. We found mainly risk scores derived from the well-known HEART score, which were validated for prehospital usage. The results tend to be very promising; for identifying high-risk patients for NSTE-ACS, but especially for excluding ACS for low-risk patients. Compared to a previous systematic review by Fanaroff et al.[42] in 2015, in-hospital use of the original HEART score shows higher predictive value for high-risk patients in diagnosing ACS, and a similar strong negative predictive value is found for patients classified as low-risk. The better predictive value for the high-risk group in the original, in-hospital HEART score can be explained by the presence of more objective information and the longer time between the onset of symptoms in relation to blood analyses. In this patient population there exists a considerable probability that the in-hospital troponin value will be positive more often compared to the prehospital setting as explained before.

In the present analysis, the performance of combined risk scores in the primary care setting were lower as compared with the EMS setting. This is due the fact that different scores were used, and in none a POC troponin. In addition, important differences exist in patient selection and prevalence of ACS, which also could have impacted the results. Finally, the performance of diagnostic tools (in particular ECG) and combined risk scores depends on the experience and support of each individual physician. In primary care there are large difference between regions and countries to what extent additional diagnostic tools are used or necessary. In particular in remote areas with no nearby hospitals, the implementation of combined risk scores could improve acute medical care for suspect ACS patients.

When evaluating a patient with suspected ACS, life-threatening diagnoses such as aortic dissection and pulmonary embolism should be part of the clinical evaluation and differential diagnosis. These alternative diagnoses are often associated with increased levels of biomarkers or ECG changes as well. So, the question is if the prehospital tested biomarkers and combined risk score are sufficient to

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also rule out chest pain due to aortic dissection or pulmonary embolism. Our meta-analysis included 3 studies that reported life-threatening diagnoses in the different risk groups. In the group of patients classified as low-risk for ACS, one pulmonary embolism and one acute pancreatitis occurred. No information regarding aortic dissections was available. These data indicate that although a combined risk score is a very strong predictor for excluding ACS, the possibility for a severe alternative diagnosis should not be ignored.

Further perspective and upcoming studies

Before using these new diagnostic tools in routine practice, more data on safety, efficacy and, performance are necessary. More data are needed when these combined risk scores are used to determine a different triage and/or treatment strategy. Some up-coming studies that will examine whether this early risk stratification and associated referral decisions based on combined risk score are feasible, are the preHEART3 study (Trialregister.nl, NL7866), PRESTO study[43], ARTICA trial[44], FamouS Triage 3 study[45] and POPular HEART (Clinicaltrials.gov, NCT04851418).

Study limitations

Although we included only prospective studies to improve the overall study quality, this analysis suffers from several limitations inherent to this study design and patient population.

An important limitation of the study was verification bias. There was a variable definition for ACS between the studies, for example not all studies included unstable angina pectoris and in some studies some patients were included who developed a STEMI. Although this reflects daily practice, it could have impacted the results.

In the analysis of some variables, heterogeneity was considerable in our meta-analysis (e.g., cardiovascular risk factors and POC-Troponin measurement in the EMS setting). In POC-Troponin,

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heterogeneity is the result of variation in thresholds between studies. Although this is expected in systematic reviews of diagnostic test accuracy, it is possible this influences our main findings.[46] However, given the small contribution we do not think that the heterogeneity in our meta-analysis alternates the influence of stand-alone risk factors in early risk stratification for ACS.

Another limitation is the lack of well-described patient and additional diagnostic tools data. For example, only 1 of 3 studies described their definition of an ischemic ECG. Therefore, it is difficult to draw conclusions about the diagnostic value of the ECG in this particular population in the prehospital setting. On the other hand, these limited available data is not uncommon for these kind of studies. It is the result of the acute setting in which these patients presents themselves to the GP or the EMS.

Additionally, many of the diagnostic tools were examined in a limited number of studies and study populations. Given the large differences in patient presentation and prevalence of ACS, the performance of the different diagnostic tools could differ when used in a clinical setting or population with different prevalence of ACS.

Finally, not for all diagnostic tools information on false negative results and alternative life-threatening diagnoses were given. Before changing triage and consequent treatment based on these tools, more prospective data are necessary to assess the safety in these rare but important selection of patients.

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Conclusion

In the prehospital setting several new diagnostic tools are available which could improve risk stratification, triage and, early treatment in suspected ACS patients. On-site assessment of biomarkers and combined risk scores derived from the HEART score showed to be strong predictors for diagnosing patients with a NSTE-ACS and to identify patients without a NSTE-ACS in the EMS setting **(Figure 1)**. These promising results underline the importance of new prospective studies in this field.

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J.D: planning, conducting, data collection, reporting, and check, statistical analysis, article writing, submission

J.Z.: data collection and reporting, statistical analysis, article writing

A.K.: article writing

G.S.: article writing

P.H.: article writing

P.T.: article writing

L.D.: article writing

M.V.: planning, conducting, date collection, reporting, and check, statistical check, article writing

P.V.: planning, conducting, data collection and reporting, article writing, submission

The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted

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Conflict of interest statement:

None.

Legends

Figure 1 is the graphical abstract. It shows the indicative positive and negative likelihood ratio for each element and how it compares to its predictive value.

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References

1. Hoorweg BB, Willemsen RT, Cleef LE, et al. Frequency of chest pain in primary care, diagnostic tests performed and final diagnoses. *Heart* 2017;103(21):1727-32. doi: 10.1136/heartjnl-2016-310905 [published Online First: 2017/06/22]
2. Pedersen CK, Stengaard C, Friesgaard K, et al. Chest pain in the ambulance; prevalence, causes and outcome - a retrospective cohort study. *Scand J Trauma Resusc Emerg Med* 2019;27(1):84. doi: 10.1186/s13049-019-0659-6 [published Online First: 2019/08/30]
3. Thang ND, Karlson BW, Bergman B, et al. Characteristics of and outcome for patients with chest pain in relation to transport by the emergency medical services in a 20-year perspective. *Am J Emerg Med* 2012;30(9):1788-95. doi: 10.1016/j.ajem.2012.02.014 [published Online First: 2012/05/29]
4. Wibring K, Lingman M, Herlitz J, et al. Prehospital stratification in acute chest pain patient into high risk and low risk by emergency medical service: a prospective cohort study. *BMJ Open* 2021;11(4):e044938. doi: 10.1136/bmjopen-2020-044938 [published Online First: 2021/04/17]
5. Backus BE, Six AJ, Kelder JC, et al. A prospective validation of the HEART score for chest pain patients at the emergency department. *Int J Cardiol* 2013;168(3):2153-8. doi: 10.1016/j.ijcard.2013.01.255
6. Antman EM, Cohen M, Bernink PJ, et al. The TIMI risk score for unstable angina/non-ST elevation MI: A method for prognostication and therapeutic decision making. *JAMA* 2000;284(7):835-42. doi: 10.1001/jama.284.7.835
7. Harskamp R, van Peet P, Bont J, et al. The conundrum of acute chest pain in general practice: a nationwide survey in The Netherlands. *BJGP Open* 2018;2(4):bjgpopen18X101619. doi: 10.3399/bjgpopen18X101619
8. Collet JP, Thiele H, Barbato E, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Rev Esp Cardiol (Engl Ed)* 2021;74(6):544. doi: 10.1016/j.rec.2021.05.002 [published Online First: 2021/05/23]
9. Bainey KR, Gafni A, Rao-Melacini P, et al. The cost implications of an early versus delayed invasive strategy in Acute Coronary Syndromes: the TIMACS study. *J Med Econ* 2014;17(6):415-22. doi: 10.3111/13696998.2014.911184 [published Online First: 2014/04/08]
10. Nieuwets A, Poldervaart JM, Reitsma JB, et al. Medical consumption compared for TIMI and HEART score in chest pain patients at the emergency department: a retrospective cost analysis. *BMJ Open* 2016;6(6):e010694. doi: 10.1136/bmjopen-2015-010694 [published Online First: 2016/06/18]
11. de Koning E, Biersteker TE, Beeres S, et al. Prehospital triage of patients with acute cardiac complaints: study protocol of HART-c, a multicentre prospective study. *BMJ Open* 2021;11(2):e041553. doi: 10.1136/bmjopen-2020-041553 [published Online First: 2021/02/14]
12. Goodacre S, Cross E, Arnold J, et al. The health care burden of acute chest pain. *Heart* 2005;91(2):229-30. doi: 10.1136/hrt.2003.027599
13. Kienbacher CL, Fuhrmann V, van Tulder R, et al. Impact of more conservative European Society of Cardiology guidelines on the management of patients with acute chest pain. *Int J Clin Pract* 2021;75(6):e14133. doi: 10.1111/ijcp.14133 [published Online First: 2021/03/09]

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14. Boubaker H, Grissa MH, Beltaief K, et al. A new score for the diagnosis of acute coronary syndrome in acute chest pain with non-diagnostic ECG and normal troponin. *Emerg Med J* 2015;32(10):764-8. doi: 10.1136/emermed-2013-203151 [published Online First: 2015/01/07]
15. Stopyra JP, Snavely AC, Smith LM, et al. Prehospital use of a modified HEART Pathway and point-of-care troponin to predict cardiovascular events. *PLoS One* 2020;15(10):e0239460. doi: 10.1371/journal.pone.0239460 [published Online First: 2020/10/08]
16. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Accuracy of pre-hospital HEART score risk classification using point of care versus high sensitive troponin in suspected NSTE-ACS. *Am J Emerg Med* 2020;38(8):1616-20. doi: 10.1016/j.ajem.2019.158448 [published Online First: 2019/11/09]
17. Rasmussen MB, Stengaard C, Sorensen JT, et al. Predictive value of routine point-of-care cardiac troponin T measurement for prehospital diagnosis and risk-stratification in patients with suspected acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;8(4):299-308. doi: 10.1177/2048872617745893 [published Online First: 2017/12/05]
18. Whiting PF, Weswood ME, Rutjes AW, et al. Evaluation of QUADAS, a tool for the quality assessment of diagnostic accuracy studies. *BMC Med Res Methodol* 2006;6:9. doi: 10.1186/1471-2288-6-9 [published Online First: 2006/03/08]
19. McGee S. Simplifying likelihood ratios. *J Gen Intern Med* 2002;17(8):646-9. doi: 10.1046/j.1525-1497.2002.10750.x
20. Honest H, Khan KS. Reporting of measures of accuracy in systematic reviews of diagnostic literature. *BMC Health Serv Res* 2002;2:4. doi: 10.1186/1472-6963-2-4
21. Grimes DA, Schulz KF. Refining clinical diagnosis with likelihood ratios. *Lancet* 2005;365(9469):1500-5. doi: 10.1016/S0140-6736(05)66422-7
22. Shim SR, Kim SJ, Lee J. Diagnostic test accuracy: application and practice using R software. *Epidemiol Health* 2019;41:e2019007. doi: 10.4178/epih.e2019007
23. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21(11):1539-58. doi: 10.1002/sim.1186
24. Anroedh SS, Kardys I, Akkerhuis KM, et al. e-Transmission of ECGs for expert consultation results in improved triage and treatment of patients with acute ischaemic chest pain by ambulance paramedics. *Neth Heart J* 2018;26(11):562-71. doi: 10.1007/s12471-018-1187-0 [published Online First: 2018/10/26]
25. Svensson L, Isaksson L, Axelsson C, et al. Predictors of myocardial damage prior to hospital admission among patients with acute chest pain or other symptoms raising a suspicion of acute coronary syndrome. *Coron Artery Dis* 2003;14(3):225-31. doi: 10.1097/01.mca.0000063503.13456.0d [published Online First: 2003/04/19]
26. Bruins Slot MH, Rutten FH, van der Heijden GJ, et al. Diagnostic value of a heart-type fatty acid-binding protein (H-FABP) bedside test in suspected acute coronary syndrome in primary care. *Int J Cardiol* 2013;168(2):1485-9. doi: 10.1016/j.ijcard.2012.12.050 [published Online First: 2013/01/22]
27. Andersson PO, Karlsson JE, Landberg E, et al. Consequences of high-sensitivity troponin T testing applied in a primary care population with chest pain compared with a commercially available point-of-care troponin T analysis: an observational prospective study. *BMC Res Notes* 2015;8:210. doi: 10.1186/s13104-015-1174-0 [published Online First: 2015/06/04]
28. Willemsen RT, Winkens B, Kietselaer BL, et al. Evaluating possible acute coronary syndrome in primary care: the value of signs, symptoms, and plasma heart-type fatty acid-binding protein (H-FABP). A diagnostic study. *BJGP Open* 2019;3(3) doi: 10.3399/bjgpopen19X101652 [published Online First: 2019/10/04]
29. Nilsson S, Andersson PO, Borgquist L, et al. Point-of-Care Troponin T Testing in the Management of Patients with Chest Pain in the Swedish Primary Care. *Int J Family Med* 2013;2013:532093. doi: 10.1155/2013/532093 [published Online First: 2013/02/01]
30. Stengaard C, Sorensen JT, Ladefoged SA, et al. Quantitative point-of-care troponin T measurement for diagnosis and prognosis in patients with a suspected acute myocardial infarction. *Am J*

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Cardiol 2013;112(9):1361-6. doi: 10.1016/j.amjcard.2013.06.026 [published Online First: 2013/08/21]

31. Ishak M, Ali D, Fokkert MJ, et al. Fast assessment and management of chest pain patients without ST-elevation in the pre-hospital gateway (FamouS Triage): ruling out a myocardial infarction at home with the modified HEART score. *Eur Heart J Acute Cardiovasc Care* 2018;7(2):102-10. doi: 10.1177/2048872616687116 [published Online First: 2017/01/14]
32. van Dongen DN, Tolsma RT, Fokkert MJ, et al. Pre-hospital risk assessment in suspected non-ST-elevation acute coronary syndrome: A prospective observational study. *Eur Heart J Acute Cardiovasc Care* 2020;9(1_suppl):5-12. doi: 10.1177/2048872618813846 [published Online First: 2018/11/24]
33. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Value of Prehospital Troponin Assessment in Suspected Non-ST-Elevation Acute Coronary Syndrome. *Am J Cardiol* 2018;122(10):1610-16. doi: 10.1016/j.amjcard.2018.07.037 [published Online First: 2018/09/29]
34. Schols AMR, Willemsen RTA, Bonten TN, et al. A Nationwide Flash-Mob Study for Suspected Acute Coronary Syndrome. *Ann Fam Med* 2019;17(4):296-303. doi: 10.1370/afm.2401 [published Online First: 2019/07/10]
35. Sagel DC, Vlaar PJ, Waardenburg I, et al. Pre-hospital risk stratification in patients with undifferentiated chest pain. *Submitted 2021*
36. Collet JP, Thiele H, Barbato E, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Eur Heart J* 2021;42(14):1289-367. doi: 10.1093/eurheartj/ehaa575 [published Online First: 2020/08/30]
37. Boeddinghaus J, Nestelberger T, Koechlin L, et al. Early Diagnosis of Myocardial Infarction With Point-of-Care High-Sensitivity Cardiac Troponin I. *J Am Coll Cardiol* 2020;75(10):1111-24. doi: 10.1016/j.jacc.2019.12.065
38. Dekker MS, Mosterd A, van 't Hof AW, et al. Novel biochemical markers in suspected acute coronary syndrome: systematic review and critical appraisal. *Heart* 2010;96(13):1001-10. doi: 10.1136/hrt.2009.189886
39. Lippi G, Mattiuzzi C, Cervellin G. Critical review and meta-analysis on the combination of heart-type fatty acid binding protein (H-FABP) and troponin for early diagnosis of acute myocardial infarction. *Clin Biochem* 2013;46(1-2):26-30. doi: 10.1016/j.clinbiochem.2012.10.016
40. Bivona G, Agnello L, Bellia C, et al. Diagnostic and prognostic value of H-FABP in acute coronary syndrome: Still evidence to bring. *Clin Biochem* 2018;58:1-4. doi: 10.1016/j.clinbiochem.2018.04.021
41. Bruins Slot MH, van der Heijden GJ, Stelpstra SD, et al. Point-of-care tests in suspected acute myocardial infarction: a systematic review. *Int J Cardiol* 2013;168(6):5355-62. doi: 10.1016/j.ijcard.2013.08.002 [published Online First: 2013/09/05]
42. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does This Patient With Chest Pain Have Acute Coronary Syndrome?: The Rational Clinical Examination Systematic Review. *JAMA* 2015;314(18):1955-65. doi: 10.1001/jama.2015.12735
43. Alghamdi A, Cook E, Carlton E, et al. PRe-hospital Evaluation of Sensitive TrOponin (PRESTO) Study: multicentre prospective diagnostic accuracy study protocol. *BMJ Open* 2019;9(10):e032834. doi: 10.1136/bmjopen-2019-032834
44. Aarts GWA, Camaro C, van Geuns RJ, et al. Acute rule-out of non-ST-segment elevation acute coronary syndrome in the (pre)hospital setting by HEART score assessment and a single point-of-care troponin: rationale and design of the ARTICA randomised trial. *BMJ Open* 2020;10(2):e034403. doi: 10.1136/bmjopen-2019-034403
45. van Dongen DN, Tolsma RT, Fokkert MJ, et al. Referral decisions based on a prehospital HEART score in suspected non-ST-elevation acute coronary syndrome: design of the FamouS Triage 3 study. *Future Cardiol* 2020;16(4):217-26. doi: 10.2217/fca-2019-0030
46. P M, C G, JJ D, et al. Chapter 10: Analysing and Presenting Results. In: JJ D, PM B, C G, eds. Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy Version 10. http://srdta.cochrane.org/: The Cochrane Collaboration 2010.

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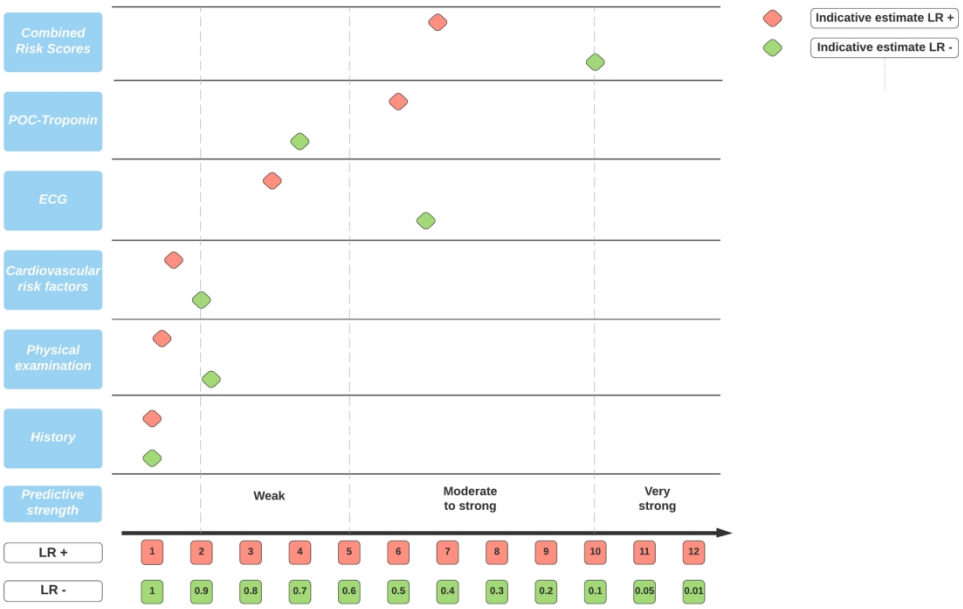


Figure 1

161x104mm (300 x 300 DPI)

Supplemental material

Supplement to: Demandt JPA, Zelis JM, Koks A, et al. Prehospital risk assessment in patients suspected of Non-ST-segment elevation Acute Coronary Syndrome – A systematic review and meta-analysis

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eAppendix 1. Full search

eAppendix 2. Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool

eAppendix 3. Study Characteristics

eAppendix 4. Results of individual studies of symptoms

eAppendix 5. Results of individual studies of risk factors

eAppendix 6. Results of individual studies of electrocardiogram findings

eAppendix 7. Results of individual studies of biomarkers

eAppendix 8. Results of individual studies of Combined Risk Scores

eAppendix 9. Results of individual studies of general practitioner decision making

eAppendix 10. Forest plots

eFigure 1. Flow diagram of study inclusion

eTable 1. Performance of symptoms for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 2. Performance of physical examination for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 3. Performance of risk factors in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 4. Performance of risk factors in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

eTable 5. Additional information about Point-of-Care (POC) analyzers

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eAppendix 1. Full search

Search strategy in MEDLINE:
(ambulance OR pre-hospital OR prehospital OR triage OR paramedics OR ems) AND (ecg OR troponin OR "risk scores" OR POC) AND (acute coronary syndrome OR acs OR nstemi OR "chest pain" OR (acute coronary syndrome [MeSH Terms]) OR (chest pain[MeSH Terms])).

Specific for primary care studies, we performed a second search:
("general practitioner" OR "primary care" OR outpatient) AND (ecg OR troponin OR "risk scores" OR POC) AND (acute coronary syndrome OR acs OR nstemi OR "chest pain" OR (acute coronary syndrome [MeSH Terms]) OR (chest pain[MeSH Terms]))

For our EMBASE search, we added a limitation to exclude all MEDLINE journals and we replaced the MeSH terms with the appropriate Emtree terms.
((ambulance or pre-hospital or prehospital or triage or paramedics or ems).mp. or exp ambulance/) and (ecg or troponin or risk scores or POC).mp. and ((acute coronary syndrome or ACS or NSTEMI or chest pain).mp. or exp non st segment elevation acute coronary syndrome/ or exp acute coronary syndrome/).

In EMBASE we performed a second search as well for additional primary care studies.
(general practitioner OR primary care OR outpatient OR exp general practitioner/ OR primary health care/) and (ecg or troponin or risk scores or POC) and (acute coronary syndrome or ACS or NSTEMI or chest pain or exp non st segment elevation acute coronary syndrome/ or exp acute coronary syndrome/).

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eAppendix 2. Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS) Tool[1]

QUADAS tool: description

1. Was the spectrum of patients representative of the patients who will receive the test in practice?
   Patients presenting in a prehospital setting (EMS, GP) with suspected ACS = yes, retrospective=no

2. Is the reference standard likely to classify the target condition correctly?
   If reference standard in hospital adjudication of ACS, especially NSTEMI or unstable AP; if reference standard is Major Adverse Cardiac Events (MACE), including at least acute myocardial infarction (AMI) =yes

3. Is the time period between reference standard and index test short enough to be reasonably sure that the target condition did not change between the two tests?
   Adjudicated diagnosis at dismissal. MACE within 6 weeks=yes

4. Did the whole sample or a random selection of the sample, receive verification using a reference standard of diagnosis?

5. Did patients receive the same reference standard irrespective of the index test result?

6. Was the reference standard independent of the index test (i.e., the index test did not form part of the reference standard)?

7. Were the reference standard results interpreted without knowledge of the results of the index test? (Index test results blinded)

8. Were the index test results interpreted without knowledge of the results of the reference standard? (Reference standard results blinded)

9. Were the same clinical data available when test results were interpreted as would be available when the test is used in practice?
   When the test executor had as much info as in clinical practice=yes

10. Were uninterpretable/intermediate test results reported?
    Not reported, numbers are correct=yes

11. Were withdrawals from the study explained?
    Not reported, numbers are correct=yes

Did the study provide a clear definition of what was to be considered a 'positive' result?

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QUADAS tool: judgement

Study	1	2	3	4	5	6	7	8	9	10	11	A
Van Dongen 2018	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Van Dongen 2020	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Ishak 2018	Y	Y	Y	Y	Y	N	U	U	Y	N	N	Y
Van Dongen 2020	Y	Y	Y	Y	Y	N	U	Y	Y	Y	Y	Y
Bruins Slot 2013	Y	Y	Y	N	Y	Y	Y	Y	Y	N	Y	Y
Andersson 2015	Y	Y	Y	Y	Y	N	Y	Y	Y	N	Y	Y
Schols 2019	Y	Y	Y	Y	U	Y	Y	Y	Y	Y	Y	U
Willemsen 2019	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Nilsson 2013	Y	Y	Y	Y	Y	N	Y	Y	Y	N	Y	Y
Anroedh 2018	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	-
Svensson 2003	Y	Y	Y	Y	Y	N	U	U	Y	N	N	Y
Stopyra 2020	Y	Y	Y	Y	Y	N	U	Y	Y	N	N	Y
Stengaard 2013	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y
Sagel 2021	Y	Y	Y	Y	Y	N	U	Y	Y	N	Y	Y
Rasmussen 2019	Y	Y	U	Y	Y	N	N	Y	Y	N	Y	Y

Y = yes; N = no; U = unclear

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eAppendix 3.

eAppendix 3. Study Characteristics					
Source	Study size	Incidence of ACS, No (%)	Clinical Setting	Diagnostic tests	Reference standard
Van Dongen et al.,[6] 2018	700	116 (17)	EMS	Male DM Obesity Family history of CAD Current smoker Previous AMI Previous PCI Previous CABG Previous TIA/Stroke Previous PAV POC- Troponin HEAR score HEART score	30-day MACE: ACS, death, PCI/CABG
Van Dongen et al.,[13] 2019	689	116 (17)	EMS	Core-lab PMHP score Hs-Troponin	30-day MACE: ACS, death, PCI/CABG
Ishak et al.,[14] 2016	1127	192 (17)	EMS	Modified HEART score	Final hospital diagnosis: NSTEMI or uAP
Van Dongen et al.,[15] 2018	700	96 (14)	EMS	HEART score	Final hospital diagnosis: NSTEMI or uAP
Bruins Slot et al.,[10] 2013	298	66 (22)	GP	H-FABP	Final hospital diagnosis: ACS
Andersson, et al.,[8] 2015	115	6 (5)	GP	Hs-Troponin POC-Troponin	Final hospital diagnosis: ACS
Schols et al.,[3] 2019	243	45 (19)	GP	Chest pain Duration of symptoms Worse with excercise Pressure pain Patients assumes cardiac origin of pain History of clinical vascular disease	Final hospital diagnosis: ACS

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				Male Pain not reproduced on palpation ECG GP immediately suspected a serious condition MHS GP probability assessment Combines MHS and GP probability assessment	
Willemsen et al.,[4] 2019	303	32 (11)	GP	H-FABP GP decision to refer CRS 1-4	Final hospital diagnosis: ACS
Nilsson et al.,[9] 2012	196	13 (7)	GP	POC-Troponin GP decision to refer GP decision to refer with POC-Troponin	Final hospital diagnosis: ACS
Anroedh et al.,[16] 2018	1421	144 (10)	EMS	ECG	Final hospital diagnosis: NSTEMI or uAP
Svensson et al.,[2] 2002	538	307 (57)	EMS	Pale Nausea Clammy AP Heart failure ECG Myoglobin CK-MB Troponin	Final hospital diagnosis: ACS
Stopyra et al.,[5] 2020	395	74 (19)	EMS	Male DM Obesity Family history of CAD Current smoker Previous AMI Previous PCI Previous CABG Previous TIA/Stroke	30-day MACE: ACS, death

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				Previous PAV Hypercholesterolemia Hypertension Prior coronary disease Heart failure PMHP score Core-lab PMHP score	
Stengaard et al.,[7] 2013	985	227 (23)	EMS	Male DM Current smoker Previous AMI Previous PCI Previous CABG Previous smoker POC-Troponin	Final hospital diagnosis: ACS
Sagel et al.,[12] 2021	435	53 (12)	EMS	preHEART score HEART score	3- and 7-day MACE: ACS or death
Rasmussen et al.,[11] 2017	17938	447 (2)	EMS	POC-Troponin	Final hospital diagnosis: NSTEMI

Abbreviations: ACS, Acute Coronary Syndrome; GP, General practitioner; EMS, Emergency Medical Services; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease; POC, Point-of-Care; HS, high sensitive; H-FABP; Heart-type Fatty Acid Binding Protein; HEART, History, ECG, Age, Risk Factors, Troponin; PMHP, prehospital modified HEART pathway; CRS, Combined Risk Scores; MHS, Marburg Heart Score; NSTEMI, non-ST-elevation myocard infarct; UAP, unstable angina pectoris; MACE, Major Adverse Cardiac Event

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eAppendix 4.

eAppendix 4. Results of individual studies of symptoms						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Svensson et al.,[2] 2006	Chestpain	536	294	213	13	16
	Dyspnoe	536	150	124	157	105
	Nausea	536	114	66	193	163
Schols et al.,[3] 2019	Chestpain	243	39	176	6	22
	Worse with exertion	188	17	69	19	83
	Chestpain feels like pressure pain	212	31	154	8	19
	Patients assumes cardiac origin of pain	242	28	108	17	89
	Symptoms duration <1h	243	39	176	6	22
	Symptoms duration (1-24h)	243	23	15	22	83
	Symptoms duration (>24h)	243	19	63	26	135
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives						

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eAppendix 5.

eAppendix 5. Results of individual studies of risk factors						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Stopyra et al.,[5] 2020	Prior coronary disease	389	37	76	37	239
	Previous PAV	395	4	17	70	304
Van Dongen et al.,[6] 2018	Previous PAV	700	11	20	105	564
Svensson et al.,[2] 2003	Previous AMI	536	160	80	147	149
Stopyra et al.,[5] 2020	Previous AMI	589	28	43	46	472
Van Dongen et al.,[6] 2018	Previous AMI	700	34	116	82	468
Stengaard et al.,[7] 2013	Previous AMI	985	82	206	149	548
Stopyra et al.,[5] 2020	Previous CABG	390	8	27	66	289
Van Dongen et al.,[6] 2018	Previous CABG	700	14	52	102	532
Stengaard et al.,[7] 2013	Previous CABG	985	21	13	210	721
Stopyra et al.,[5] 2020	Previous PCI	395	19	78	55	243
Van Dongen et al.,[6] 2018	Previous PCI	700	42	133	74	451
Stengaard et al.,[7] 2013	Previous PCI	985	73	179	158	575
Stopyra et al.,[5] 2020	Previous TIA/Stroke	395	7	34	67	287
Van Dongen et al.,[6] 2018	Previous TIA/Stroke	700	11	32	105	552
Svensson et al.,[2] 2003	Current smoker	700	30	126	86	458
Stopyra et al.,[5] 2020	Current smoker	395	16	87	58	234
Van Dongen et al.,[6] 2018	Current smoker	700	30	126	86	458
Stengaard et al.,[7] 2013	Current smoker	1084	80	120	151	733
Svensson et al.,[2] 2003	Male gender	536	181	128	126	101
Stopyra et al.,[5] 2020	Male gender	395	43	142	31	179
Van Dongen et al.,[6] 2018	Male gender	700	88	313	28	271
Stengaard et al.,[7] 2013	Male gender	985	170	413	61	341
Svensson et al.,[2] 2003	AP	536	196	115	111	114

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Stopyra et al.,[5] 2020	Hypercholesterolemia	395	26	83	48	238
Van Dongen et al.,[6] 2018	Hypercholesterolemia	700	61	214	55	370
Stengaard et al.,[7] 2013	Hypercholesterolemia	985	194	624	37	130
Svensson et al.,[2] 2003	DM	536	58	32	249	197
Stopyra et al.,[5] 2020	DM	395	29	94	45	227
Van Dongen et al.,[6] 2018	DM	700	26	87	90	497
Stengaard et al.,[7] 2013	DM	985	38	143	193	611
Svensson et al.,[2] 2003	Heart failure	536	68	57	239	172
Stopyra et al.,[5] 2020	Heart failure	389	14	40	59	276
Svensson et al.,[2] 2003	Hypertension	536	107	57	200	172
Stopyra et al.,[5] 2020	Hypertension	389	55	201	18	115
Van Dongen et al.,[6] 2018	Hypertension	700	66	306	50	278
Stengaard et al.,[7] 2013	Hypertension	985	133	405	98	349
Stopyra et al.,[5] 2020	Obesity	383	29	152	45	157
Van Dongen et al.,[6] 2018	Obesity	700	27	113	89	471
Stengaard et al.,[7] 2013	Previous smoker	985	69	208	162	546
Stopyra et al.,[5] 2020	Family history of CAD	395	19	78	55	243
Van Dongen et al.,[6] 2018	Family history of CAD	700	47	277	69	307
Schols et al.,[3] 2019	Gender	243	34	93	11	105
	History of clinical vascular disease	243	22	84	23	114

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease; TP, true positives; FP, false positives; FN, false negatives, TN, true negatives

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eAppendix 6.

eAppendix 6. Results of individual studies of electrocardiogram findings						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Anroedh et al.,[16] 2018	Ischemic ECG	1421	69	457	75	820
Svensson et al.,[2] 2003	Ischemic ECG	536	52	126	255	103
Schols et al.,[3] 2019	Ischemic ECG	115	17	20	6	72
Svensson et al.,[2] 2003	Q-wave	536	33	1	274	228
	ST-depression	536	129	25	178	204
	T-wave inversion	536	55	16	252	213
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives						

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eAppendix 7.

eAppendix 7. Results of individual studies of biomarkers						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Svensson et al.,[2] 2003	POC-Troponin	536	21	2	286	227
Stengaard et al.,[7] 2013	POC-Troponin	924	73	34	146	671
Stopyra et al.,[5] 2020	POC-Troponin	395	17	11	57	310
Van Dongen et al.,[6] 2018	POC-Troponin	700	43	26	73	558
Rasmussen et al.,[11] 2019	POC-Troponin	17938	640	1175	1026	15097
Andersson et al.,[8] 2015	POC-Troponin	115	2	2	4	107
Nilsson et al.,[9] 2013	POC-Troponin	128	2	3	5	118
Willemsen et al.,[4] 2019	H-FABP	291	8	8	23	252
Bruins Slot et al.,[10] 2013	H-FABP	298	26	14	40	218
Svensson et al.,[2] 2003	CK-MB	536	37	7	270	222
	Myoglobin	536	25	9	282	220
Andersson et al.,[8] 2015	Hs-Troponin	115	5	26	1	83
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; POC, Point-of-Care; H-FABP; Heart-type Fatty Acid Binding Protein; HS, high-sensitive						

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eAppendix 8.

eAppendix 8. Results of individual studies of Combined Risk Scores						
Study	Characteristic	Risk	TP	FP	FN	TN
Stopyra et al.,[5] 2020	Core-lab PMHP	High	50	25		
		Intermediate	24	172		
		Low	0	172		
	PMHP	High	17	11		
		Intermediate	50	193		
		Low	7	117		
Sagel et al.,[12] 2021	preHEART	High	55	56		
		Intermediate	117	797		
		Low	4	614		
	HEART	High	67	122		
		Intermediate	51	648		
		Low	5	315		
Ishak et al.,[14] 2018	Modified HEART	High	116	114		
		Intermediate	76	418		
		Low	0	403		
Willemsen et al.,[4] 2019	CRS 1	High - Low	28	130	4	141
	CRS 2	High - Low	26	130	6	141
	CRS 3	High - low	25	121	7	150
	CRS 4	High - Low	22	120	10	151
Schols et al.,[3] 2019	MHS	Cut-off value 2	27	84	9	66
	MHS	Cut- off value 1	34	126	2	24
	GP probability assessment	High - Low	36	116	6	82
	Combined MHS & GP probability assessment	High - Low	36	115	0	35
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; CRS, Combined Risk Scores; HEAR, History, ECG, Age, Risk Factors; POC, Point-of-Care; H-FABP, Heart-type Fatty Acid Binding Protein; MHS, Marburg Heart Score; GP, general practitioner						

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eAppendix 9.

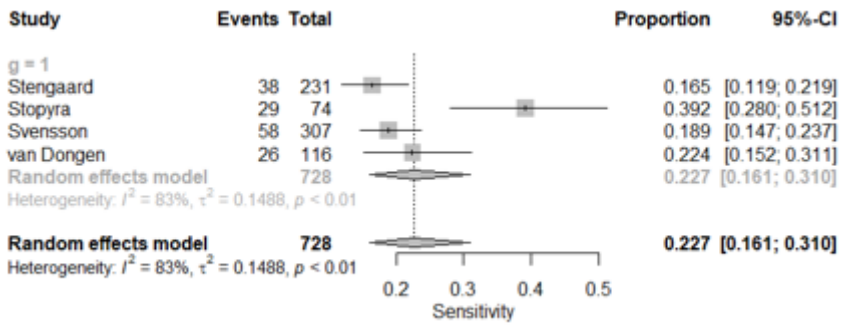
eAppendix 9. Results of individual studies of general practitioner decision making						
Study	Characteristic	Total no. patients	TP	FP	FN	TN
Willemsen et al.,[4] 2019	GP decision to refer	303	24	88	8	183
Nilsson et al.,[9] 2013	GP decision to refer	68	6	23	0	39
	GP decision to refer with POC-Troponin	128	5	27	2	94
Schols et al.,[3] 2019	GP immediately suspected a serious condition	243	26	94	19	104
Abbreviations: TP, true positives; FP, false positives; FN, false negatives, TN, true negatives; GP, General Practitioner; POC, Point-of-Care						

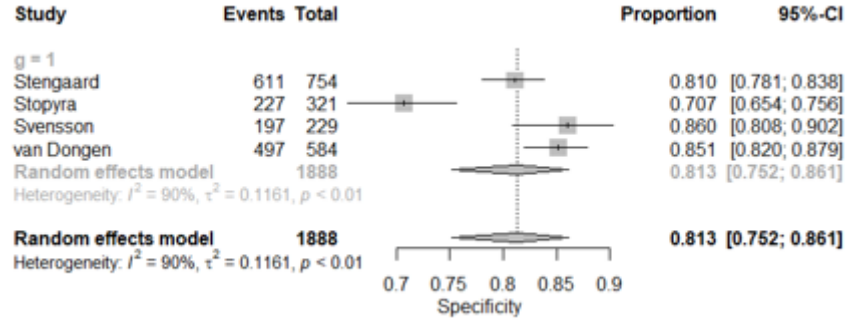
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eAppendix 10. Forest plots

Diabetes Mellitus

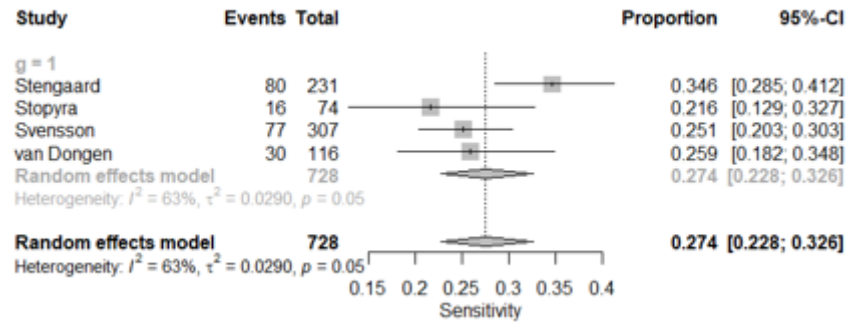


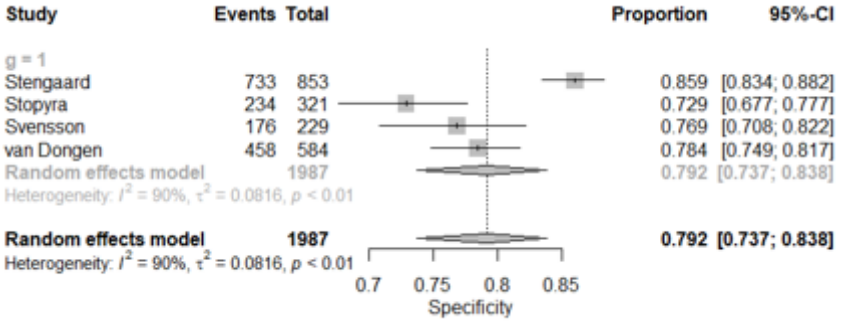


G=1 = Emergency Medical Services setting

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Current smoker

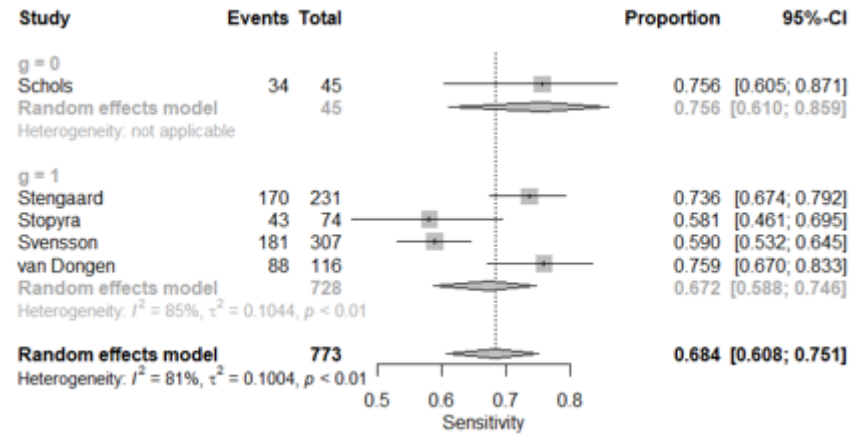


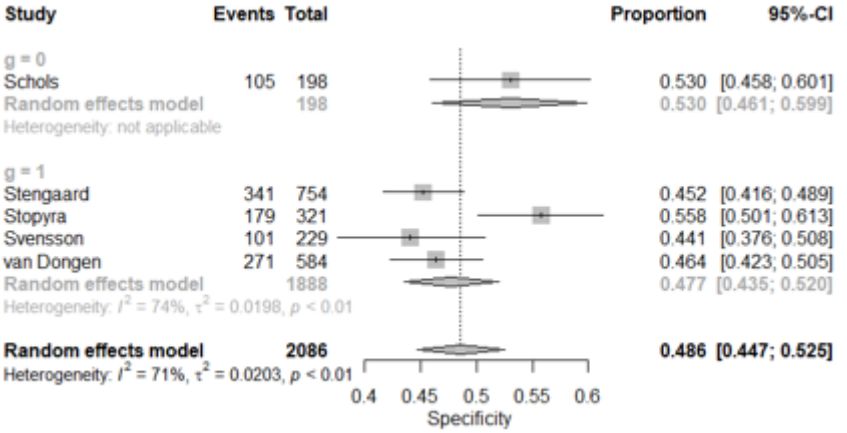


G=1 = Emergency Medical Services setting

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Male gender

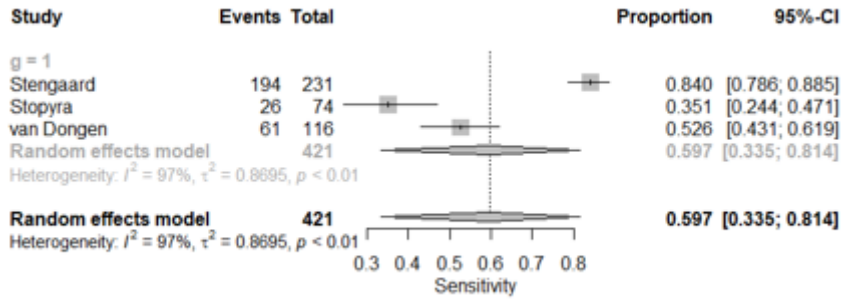


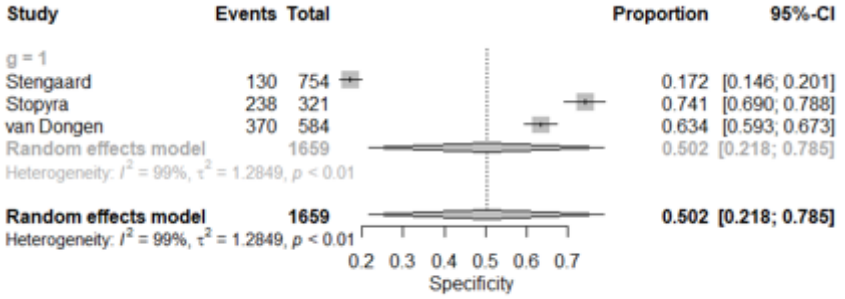


G=1 = Emergency Medical Services setting; G=-0 = General Practitioner setting

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Hypercholesterolemia

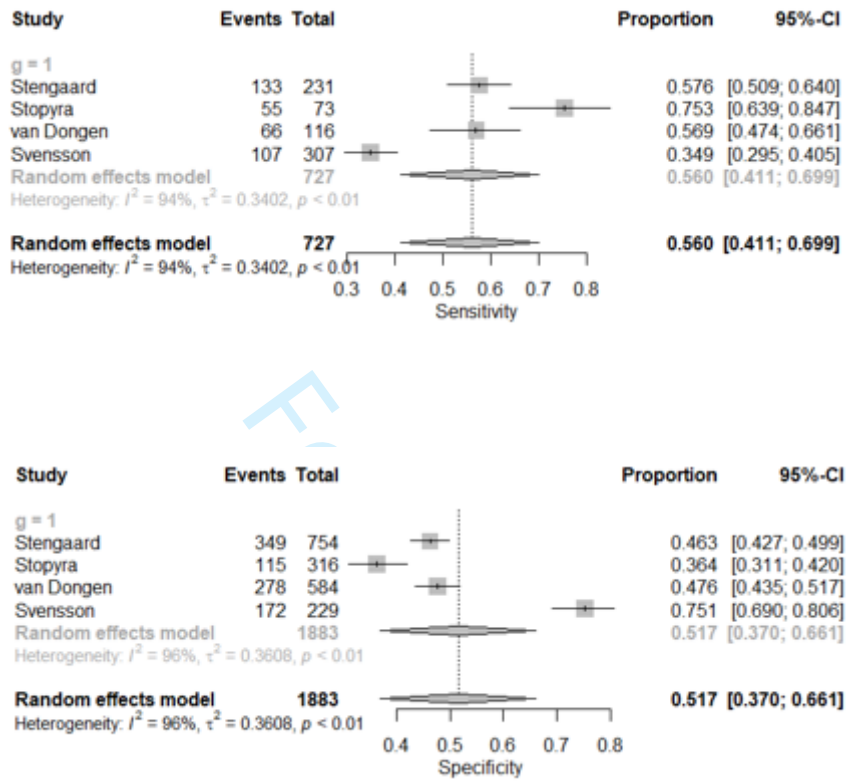




G=1 = Emergency Medical Services setting

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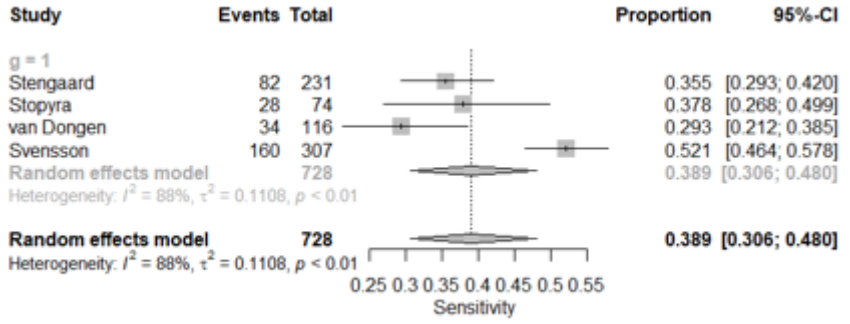
Hypertension

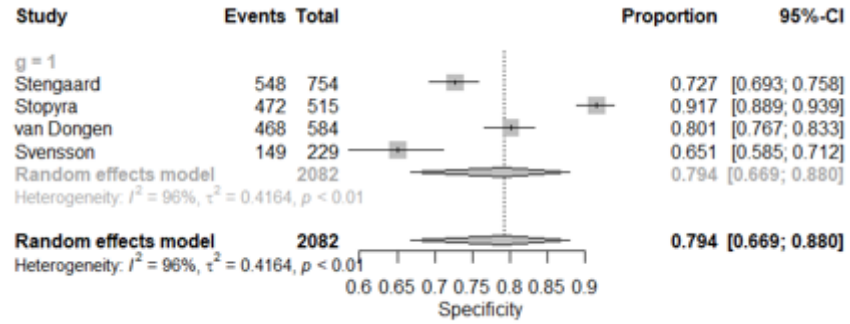


G=1 = Emergency Medical Services setting

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Previous AMI



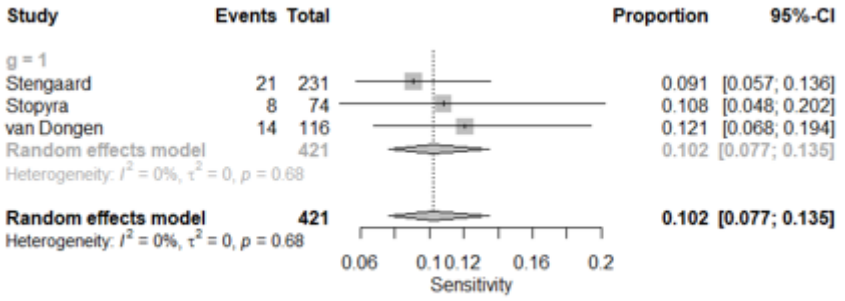


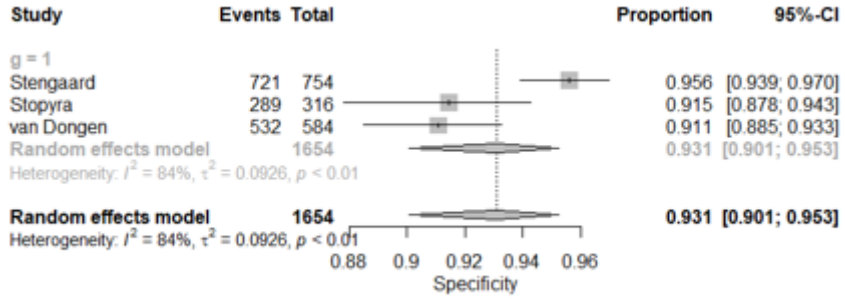
G=1 = Emergency Medical Services setting

AMI = Acute Myocardial Infarction

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Previous CABG



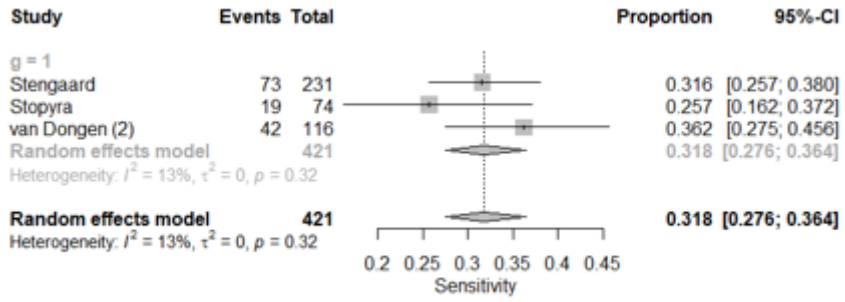


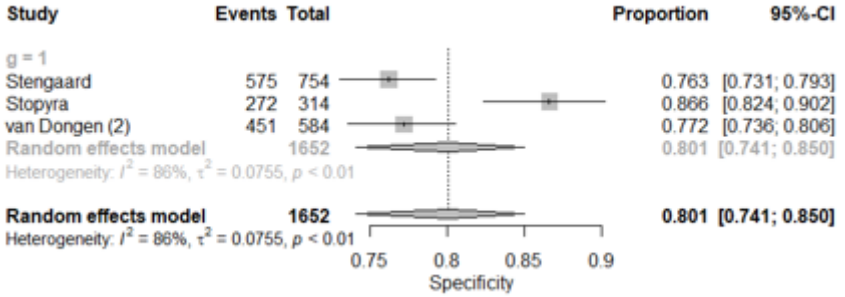
G=1 = Emergency Medical Services setting

CABG = Coronary Artery Bypass Graft

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Previous PCI





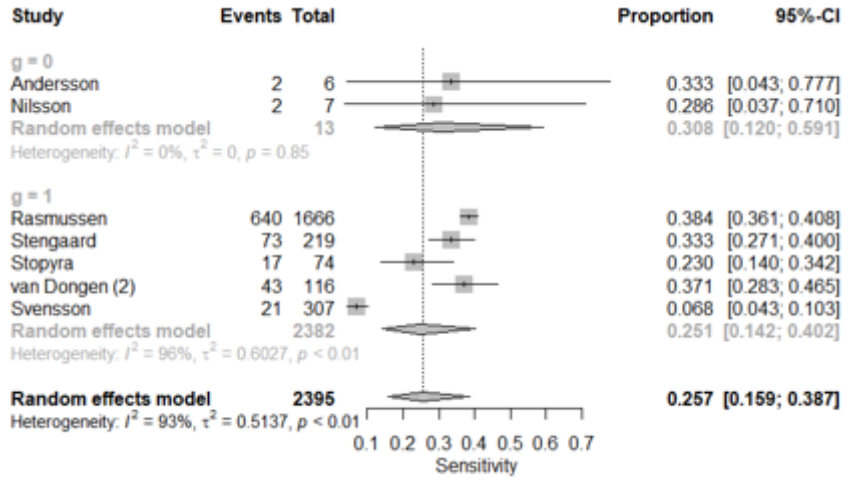
G=1 = Emergency Medical Services setting

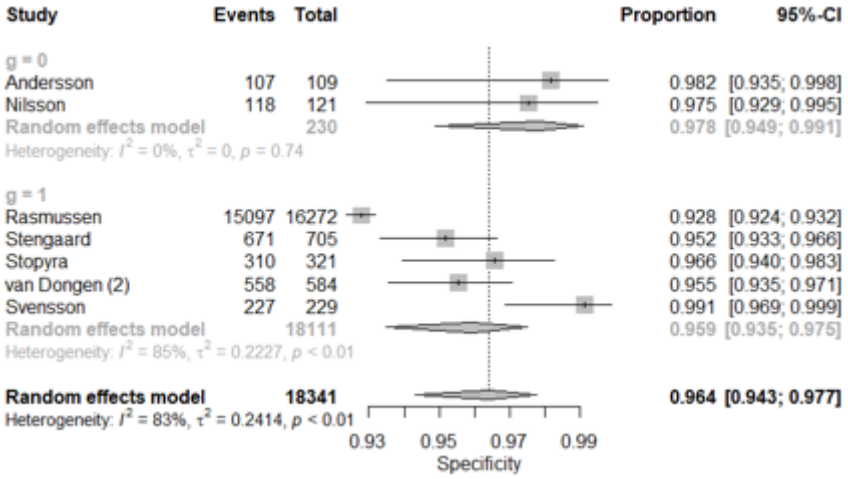
PCI = Percutaneous Coronary Intervention

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POC Troponin



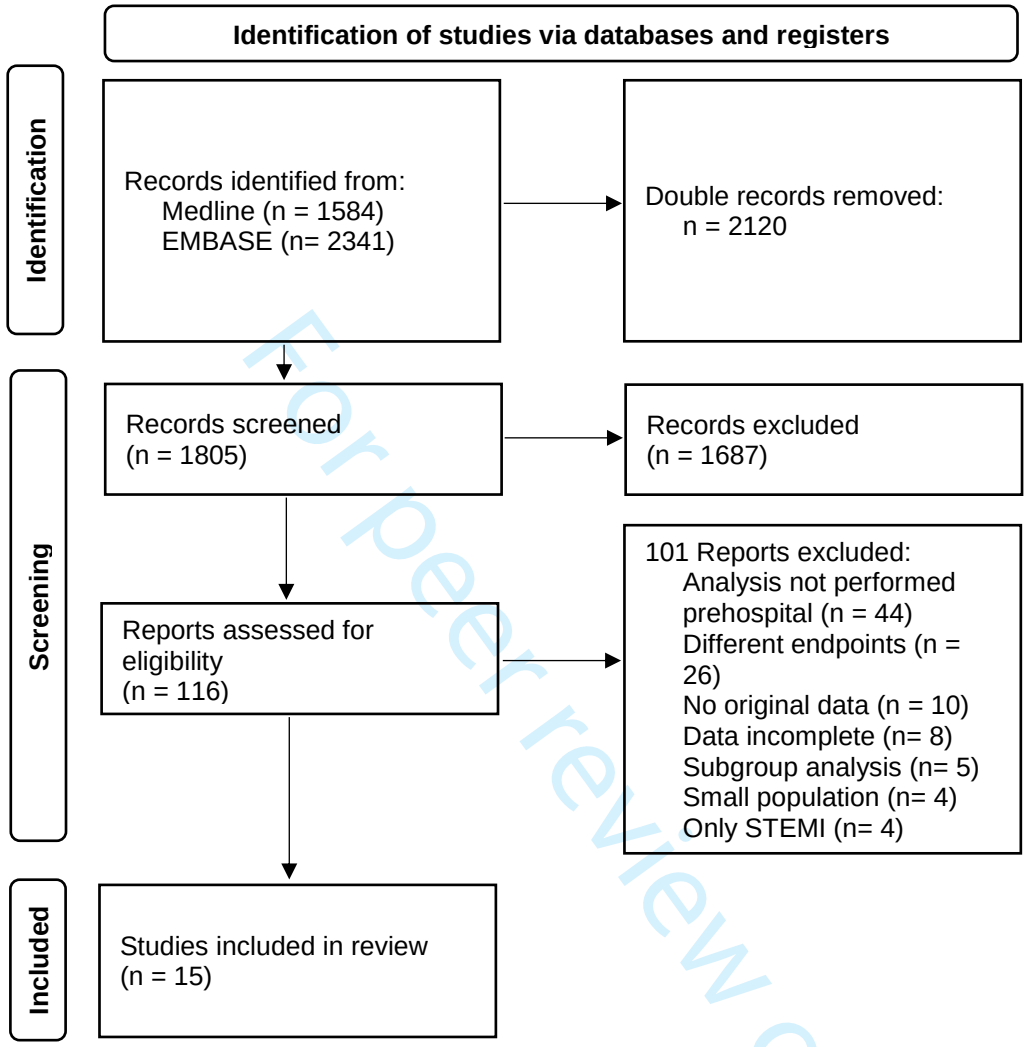


G=1 = Emergency Medical Services setting; G=0 = General Practitioner setting

POC = Point-of-Care

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eFigure 1. Flow diagram of study inclusion



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eTable 1.

eTable 1. Performance of symptoms for risk assessment in non-ST-elevation Acute Coronary Syndrome										
Symptom	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Chestpain[2]	EMS	1	538	96 (93-98)	-	7 (4-11)	-	1.03 (0.98-1.07)	0.61 (0.58-0.63)	1.70 (0.80-3.61)
Dyspnoe [2]	EMS	1	536	49 (43-55)	-	46 (39-53)	-	0.90 (0.76-1.07)	1.12 (0.94-1.32)	0.81 (0.57-1.14)
Nausea [2]	EMS	1	536	37 (32-43)	-	71 (65-77)	-	1.29 (1.00-1.67)	0.88 (0.68-1.14)	1.46 (1.01-2.11)
Chestpain [3]	GP	1	243	87 (73-95)	-	11 (7-16)	-	0,98 (0.90-1.05)	1.29 (1.11-1.29)	0.81 (0.31-2.14)
Worse with exertion [3]	GP	1	212	47 (30-65)	-	55 (46-63)	-	1.0 (0.79-1.38)	0.97 (0.73-1.28)	1.08 (0.52-2.23)
Chestpain feels like pressure [3]	GP	1	212	79 (64-91)	-	11 (7-17)	-	0.89 (0.81-0.98)	1.87 (1.70-2.05)	0.48 (0.19-1.19)
Patient assumes cardiac origin of pain [3]	GP	1	242	62 (47-76)	-	45 (38-52)	-	1.13 (0.94-1.37)	0.84 (0.69-1.01)	1.36 (0.70-2.64)
Symptoms duration <1h [3]	GP	1	243	87 (73-95)	-	11 (7-16)	-	0.98 (0.90-1.05)	1.2 (1.11-1.29)	0.81 (0.31-2.14)

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Symptoms duration (1-24h) [3]	GP	1	243	51 (36-66)	-	42 (35-49)	-	0.88 (0.72-1.08)	1.16 (0.95-1.43)	0.75 (0.39-1.44)
Symptoms duration (>24h)[3]	GP	1	243	42 (28 58)	-	68 (61-75)	-	1.33 (0.97-1.82)	0.85 (0.62-1.16)	1.57 (0.81-3.04)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 2

eTable 2. Performance of physical examination for risk assessment in non-ST-elevation Acute Coronary Syndrome

Test	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Pale [2]	EMS	1	536	60 (54-65)	-	49 (42-56)	-	1.17 (1.00-1.38)	0.82 (0.70-0.96)	1.43 (1.01-2.02)
Clammy [2]	EMS	1	536	37 (32-43)	-	68 (62-74)	-	1.16 (0.21-1.49)	0.92 (0.72-1.18)	1.26 (0.88-1.81)
Pain not reproducible by palpation [3]	GP	1	210	92 (79-98)	-	16 (11-23)	-	1.10 (1.02-1.20)	0.47 (0.43-0.51)	2.35 (0.68-8.16)
Cardiac murmur [4]	GP	1	250	4 (0-19)	-	91 (86-94)	-	0.41 (0.08-1.91)	1.06 (0.23-4.89)	0.39 (0.05-3.03)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 3

eTable 3. Performance of risk factors in EMS setting for risk assessment in non-ST-elevation Acute Coronary Syndrome										
Risk factor	Setting	Studies	Patients	Sensitivity % (95-CI)	I^2, %[a]	Specificity % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Prior coronary disease [5]	EMS	1	389	50 (38-62)	-	76 (71-80)	-	2.07 (1.58-2.71)	0.66 (0.50-0.86)	3.14 (1.86-5.31)
Previous PAV [5 6]	EMS	2	1095	8 (1-16)	-	96 (92-98)	-	1.93 (0.38-5.53)	0.96 (0.37-2.70)	2.01 (0.33-6.35)
Previous AMI [2 5-7]	EMS	4	2618	39 (31-48)	88%	79 (69-88)	96%	1.84 (1.21-2.80)	0.80 (0.71-0.91)	2.32 (1.34-4.00)
Previous CABG [5-7]	EMS	3	2080	10 (8-14)	0%	93 (90-95)	84%	1.60 (1.14-2.24)	0.96 (0.93-0.99)	1.67 (1.15-2.44)
Previous PCI [5-7]	EMS	3	2080	32 (28-36)	13%	80 (74-85)	86%	1.49 (1.24-1.78)	0.87 (0.81-0.93)	1.71 (1.35-2.17)
Previous TIA/Stroke [5 6]	EMS	2	1095	9 (4-19)	-	93 (86-96)	-	1.30 (0.45-3.21)	0.98 (0.52-2.00)	1.33 (0.37-3.7)
Current smoker [2 5-7]	EMS	4	2618	27 (23-33)	63%	79 (74-84)	90%	1.29 (0.78-2.14)	0.93 (0.80-1.07)	1.39 (0.72-2.68)
Male gender [2 5-7]	EMS	4	2618	67 (59-75)	85%	48 (44-52)	74%	1.28 (1.12-1.45)	0.69 (0.53-0.90)	1.86 (1.39-2.80)

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AP [2]	EMS	1	536	64 (58-69)	-	50 (43-56)	-	1.27 (1.08-1.49)	0.73 (0.62-0.85)	1.75 (1.24-2.48)
Hypercholesterolemia [5-7]	EMS	3	2080	60 (34-81)	97%	50 (22-79)	99%	1.23 (0.93-1.61)	0.83 (0.73-0.94)	1.47 (1.04-2.10)
DM [2 5-7]	EMS	4	2618	23 (16-31)	83%	81 (75-86)	90%	1.22 (0.95-1.57)	0.96 (0.89-1.03)	1.29 (0.93-1.80)
Heart failure [2 5]	EMS	2	933	22 (11-30)	-	82 (69-91)	-	1.21 (0.65-2.49)	0.95 (0.56-1.52)	1.27 (0.57-3.2)
Hypertension [2 5-7]	EMS	4	2618	56 (41-70)	94%	52 (37-66)	96%	1.14 (1.04-1.25)	0.88 (0.81-0.95)	1.33 (1.10-1.61)
Obesity [5 6]	EMS	2	1095	29 (16-51)	-	70 (45-84)	-	0.99 (0.65-1.62)	1.00 (0.70-1.47)	0.99 (0.40-2.04)
Previous smoker [7]	EMS	1	985	30 (24-36)	-	72 (69-76)	-	1.08 (0.89-1.31)	0.97 (0.80-1.18)	1.12 (0.81-1.55)
Family history of CAD [5 6]	EMS	2	1095	35 (16-50)	-	61 (48-80)	-	0.89 (0.73-1.50)	1.07 (0.69-1.39)	0.82 (0.50-1.92)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio; DM, diabetes mellitus; AMI, acute myocardial infarction; CABG, Coronary Artery Bypass Grafting; PCI, Percutaneous Coronary Intervention; TIA; Transient ischemic attack, PAV; Peripheral artery disease; AP, angina pectoris; CAD, Coronary artery disease

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 4

eTable 4. Performance of risk factors in primary care for risk assessment in non-ST-elevation Acute Coronary Syndrome

Risk factor	Setting	Studies	Patients	Sensitivity, % (95-CI)	I^2, %[a]	Specificity, % (95-CI)	I^2, %[a]	LR+ (95-CI)	LR- (95-CI)	dOR (95-CI)
Male gender [3]	GP	1	243	77 (61-86)	-	53 (46-60)	-	1.61 (1.34-1.93)	0.46 (0.38-0.55)	3.49 (1.67-7.23)
History of clinical vascular disease [3]	GP	1	243	49 (34-64)	-	58 (50-65)	-	1.15 (0.90-1.48)	0.89 (0.69-1.14)	1.30 (0.68-2.48)

Abbreviations: 95-CI, 95% confidence interval; LR+, positive likelihood ratio; LR- negative likelihood ratio; dOR, diagnostic odds ratio

[a] When the summary measure was from less than 3 studies, I^2 was not calculated

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eTable 5.

eTable 5. Additional information about Point-of-Care (POC) analyzers							
Study	Setting	Manufacturer	Assay	Median duration (minutes)	Test result (minutes)	Succes rate (%)	Positive threshold
Van Dongen[6]	EMS	Roche Diagnostics	Troponin T	150	8 – 12	90	0.04 µg/L
Andersson [8]	GP	Roche Diagnostics	Troponin T	600	14	-	0.03 µg/L
Willemsen [4]	GP	FABPulous BV	H-FABP	-	-	-	4 µg/L
Nilsson [9]	GP	Roche Diagnostics	Troponin T	600	14	-	0.03 µg/L
Bruins Slot [10]	GP	Cardiodetect Rennesens GmbH	H-FABP	180	15	89	7 µg/L
Stopyra [5]	EMS	Abbott	Troponin I	-	8-10	-	0.08 µg/L
Svensson [2]	EMS	Cardiac Status Spectral Diagnostics Tests	Myoglobin CK-MB Troponin I	90	15	-	2.91 nmol/L 5 µg/L 0.1 µg/L
Stengaard [7]	EMS	Roche Diagnostics	Troponin T	70	12	90%	0.05 µg/L

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Rasmussen [11]	EMS	Roche Diagnostics	Troponin T	55	12	-	0.05 µg/L
Sage[12]	EMS	Abbott	Troponin I	294	8-10	-	0.34 µg/L
Abbreviations: POC, Point-of-Care; HS, high sensitive; GP, General practitioner; EMS, Emergency Medical Services							

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References

1. Whiting PF, Weswood ME, Rutjes AW, et al. Evaluation of QUADAS, a tool for the quality assessment of diagnostic accuracy studies. *BMC Med Res Methodol* 2006;6:9. doi: 10.1186/1471-2288-6-9 [published Online First: 2006/03/08]
2. Svensson L, Isaksson L, Axelsson C, et al. Predictors of myocardial damage prior to hospital admission among patients with acute chest pain or other symptoms raising a suspicion of acute coronary syndrome. *Coron Artery Dis* 2003;14(3):225-31. doi: 10.1097/01.mca.0000063503.13456.0d [published Online First: 2003/04/19]
3. Schols AMR, Willemsen RTA, Bonten TN, et al. A Nationwide Flash-Mob Study for Suspected Acute Coronary Syndrome. *Ann Fam Med* 2019;17(4):296-303. doi: 10.1370/afm.2401 [published Online First: 2019/07/10]
4. Willemsen RT, Winkens B, Kietselaer BL, et al. Evaluating possible acute coronary syndrome in primary care: the value of signs, symptoms, and plasma heart-type fatty acid-binding protein (H-FABP). A diagnostic study. *BJGP Open* 2019;3(3) doi: 10.3399/bjgpopen19X101652 [published Online First: 2019/10/04]
5. Stopyra JP, Snavely AC, Smith LM, et al. Prehospital use of a modified HEART Pathway and point-of-care troponin to predict cardiovascular events. *PLoS One* 2020;15(10):e0239460. doi: 10.1371/journal.pone.0239460 [published Online First: 2020/10/08]
6. van Dongen DN, Fokkert MJ, Tolsma RT, et al. Value of Prehospital Troponin Assessment in Suspected Non-ST-Elevation Acute Coronary Syndrome. *Am J Cardiol* 2018;122(10):1610-16. doi: 10.1016/j.amjcard.2018.07.037 [published Online First: 2018/09/29]
7. Stengaard C, Sorensen JT, Ladefoged SA, et al. Quantitative point-of-care troponin T measurement for diagnosis and prognosis in patients with a suspected acute myocardial infarction. *Am J Cardiol* 2013;112(9):1361-6. doi: 10.1016/j.amjcard.2013.06.026 [published Online First: 2013/08/21]
8. Andersson PO, Karlsson JE, Landberg E, et al. Consequences of high-sensitivity troponin T testing applied in a primary care population with chest pain compared with a commercially available point-of-care troponin T analysis: an observational prospective study. *BMC Res Notes* 2015;8:210. doi: 10.1186/s13104-015-1174-0 [published Online First: 2015/06/04]
9. Nilsson S, Andersson PO, Borgquist L, et al. Point-of-Care Troponin T Testing in the Management of Patients with Chest Pain in the Swedish Primary Care. *Int J Family Med* 2013;2013:532093. doi: 10.1155/2013/532093 [published Online First: 2013/02/01]
10. Bruins Slot MH, Rutten FH, van der Heijden GJ, et al. Diagnostic value of a heart-type fatty acid-binding protein (H-FABP) bedside test in suspected acute coronary syndrome in primary care. *Int J Cardiol* 2013;168(2):1485-9. doi: 10.1016/j.ijcard.2012.12.050 [published Online First: 2013/01/22]
11. Rasmussen MB, Stengaard C, Sorensen JT, et al. Predictive value of routine point-of-care cardiac troponin T measurement for prehospital diagnosis and risk-stratification in patients with suspected acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;8(4):299-308. doi: 10.1177/2048872617745893 [published Online First: 2017/12/05]
12. Sagel DC, Vlaar PJ, Waardenburg I, et al. Pre-hospital risk stratification in patients with undifferentiated chest pain. *Submitted 2021*

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Research Reporting Guideline checklist

Demandt JPA, Zelis JM, Koks A, et al. Prehospital Risk Assessment In Suspected Acute Coronary Syndrome Patients – A Systematic Review And Meta-analysis

PRISMA-DTA Checklist

Section/topic	#	PRISMA-DTA Checklist Item	Reported on page #
TITLE / ABSTRACT			
Title	1	Identify the report as a systematic review (+/- meta-analysis) of diagnostic test accuracy (DTA) studies.	1
Abstract	2	Abstract: See PRISMA-DTA for abstracts.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5/6
Clinical role of index test	D1	State the scientific and clinical background, including the intended use and clinical role of the index test, and if applicable, the rationale for minimally acceptable test accuracy (or minimum difference in accuracy for comparative design).	5/6
Objectives	4	Provide an explicit statement of question(s) being addressed in terms of participants, index test(s), and target condition(s).	5/6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	7
Eligibility criteria	6	Specify study characteristics (participants, setting, index test(s), reference standard(s), target condition(s), and study design) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7/8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full search strategies for all electronic databases and other sources searched, including any limits used, such that they could be repeated.	Supplementary
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Definitions for data extraction	11	Provide definitions used in data extraction and classifications of target condition(s), index test(s),	8

		reference standard(s) and other characteristics (e.g. study design, clinical setting).	
Risk of bias and applicability	12	Describe methods used for assessing risk of bias in individual studies and concerns regarding the applicability to the review question.	7
Diagnostic accuracy measures	13	State the principal diagnostic accuracy measure(s) reported (e.g. sensitivity, specificity) and state the unit of assessment (e.g. per-patient, per-lesion).	8
Synthesis of results	14	Describe methods of handling data, combining results of studies and describing variability between studies. This could include, but is not limited to: a) handling of multiple definitions of target condition. b) handling of multiple thresholds of test positivity, c) handling multiple index test readers, d) handling of indeterminate test results, e) grouping and comparing tests, f) handling of different reference standards	8

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Section/topic	#	PRISMA-DTA Checklist Item	Reported on page #
Meta-analysis	D2	Report the statistical methods used for meta-analyses, if performed.	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	-
RESULTS			
Study selection	17	Provide numbers of studies screened, assessed for eligibility, included in the review (and included in meta-analysis, if applicable) with reasons for exclusions at each stage, ideally with a flow diagram.	10/Supplementary
Study characteristics	18	For each included study provide citations and present key characteristics including: a) participant characteristics (presentation, prior testing), b) clinical setting, c) study design, d) target condition definition, e) index test, f) reference standard, g) sample size, h) funding sources	Supplementary
Risk of bias and applicability	19	Present evaluation of risk of bias and concerns regarding applicability for each study.	Supplementary
Results of individual studies	20	For each analysis in each study (e.g. unique combination of index test, reference standard, and positivity threshold) report 2x2 data (TP, FP, FN, TN) with estimates of diagnostic accuracy and confidence intervals, ideally with a forest or receiver operator characteristic (ROC) plot.	Supplementary
Synthesis of results	21	Describe test accuracy, including variability; if meta-analysis was done, include results and confidence intervals.	10-21
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression; analysis of index test: failure rates, proportion of inconclusive results, adverse events).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence.	23-26
Limitations	25	Discuss limitations from included studies (e.g. risk of bias and concerns regarding applicability) and from the review process (e.g. incomplete retrieval of identified research).	26/27
Conclusions	26	Provide a general interpretation of the results in the context of other evidence. Discuss implications for future research and clinical practice (e.g. the intended use and clinical role of the index test).	23-27
FUNDING			
Funding	27	For the systematic review, describe the sources of funding and other support and the role of the funders.	29

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